INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 4: Implementation and Empirical Results

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submitted in accordance with the requirements for the degree of

DOCTOR OF PHILOSOPHY

in the subject

COMPUTER SCIENCE

at the

UNIVERSITY OF SOUTH AFRICA

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October 2026

4 Implementation and Empirical Results

This chapter runs the design of Chapter 3 on Arbitrum One as a single case study: one chain and one observation window, one matched cohort of contracts for every same-window comparison, spender and trader cohorts for the temporal holdouts W0 and W1, and subsamples of the matched cohort for runtime against size. It reports the two single-layer walks first, their cost, their agreement in the window, their divergence, robustness checks and their holdouts (Sections 4.2 to 4.6), and then the coupled walk C-PR, the method of the thesis, under its rules and beside every other score (Section 4.7); a comparison on outcomes built from neither edge type follows (Section 4.8). It reports the revised analysis of Section 3.5.1, with amounts in US dollars and five tokens; the revision was decided after the registered results were known, so no result in this chapter is a registered result. Rules A and B and R1–R3 are applied to the revised scores, with their thresholds and wording, as a re-analysis, and the registered analysis reached the same verdicts (Appendix 8.6, Table 8.12). Every table is generated from the evaluation summaries of Section 3.13, each Kendall τ comes with its confidence interval except in the value-slope and sample-definition checks, and each PageRank is read against the raw degree it smooths.

4.1 Case study setting: Arbitrum One

Arbitrum One was chosen because it is a heavily used Ethereum rollup with mature DeFi protocols (Section 2.2), because its logs are in the public BigQuery export that the pipeline reads, and because one of its margin-trading protocols, GMX V2, records the position closes behind the trader labels. The observation window, 1 October 2023 to 30 June 2026 (33 months), spans several market regimes. The time decay is anchored at its end, T_{obs} (Figure 3.3), so the last year carries most of the edge weight in AWP and EndorseRank.

4.1.1 Cohorts

Four sets of contracts are used. Every address in them holds contract code; externally owned accounts (EOAs) enter the graphs as owners, senders and recipients and are never scored (Section 3.2).

- Matched cohort ($n = 15{,}107$). The contracts that received a non-zero `Approval` of one of the five tokens from at least three distinct owners in the window (Section 3.2). They are part of the analysis plan's cohort of 27,844 contracts, which applied the same rule to every token and left out as EOAs 1,121 of its 29,022 candidates without code and 57 with an EIP-7702 delegation designator. Membership does not depend on either graph: no allowance edge touches 523 cohort contracts (by T_{obs} every allowance granted to them or by them was zero), no transfer edge touches $2{,}670$, and 80 are touched by neither; a contract absent from a graph scores zero under that method. Every same-window alignment, difference test and benchmark is computed on this cohort.
- Spender cohorts of W0 and W1 ($n = 17{,}050$ and $18{,}152$). In W0 the scores are frozen at t_1 (31 March 2026, 23:59:59 UTC), and the cohort is every spender that holds at least one positive latest allowance of one of the five tokens at t_1 in the extracted approval data and holds contract code: 17,050 of 17,167 such spenders. In W1 the scores are frozen at T_{obs}, and the same rule gives 18,152 contract spenders. The spender cohorts are not subsets of the matched cohort, since a spender needs only one owner with a positive allowance. The scores of these contracts are tested against owner–spender approval pairs and transfer senders first seen in the three months after each freeze (Section 3.7).
- Trader cohorts ($n = 88$ and 70). The GMX V2 accounts that hold contract code, are nodes of the freeze-date graph and close at least three positions in the label window: 88 in W0, 40 of which had received an approval by the freeze, and 70 in W1, 37 of which had. Contract accounts closed 133,685 GMX V2 positions in the observation window, but only 9,809 in its last six months and 6,805 from July to September 2026 (Table 8.5 in Appendix 8.4.3). Every trader result therefore rests on few contracts and has wide intervals.

- Subsamples of the matched cohort ($n = 1{,}000$, $2{,}000$, $5{,}000$ and $10{,}000$). Deterministic SHA256 subsamples (seed `contract-benchmark-v1`), followed by the full cohort, used only for runtime against size and for the sensitivity to sample definition (Section 3.12.1). Each stage keeps the extracted edges incident to its sampled contracts, so its edge count grows with n.

At T_{obs} the allowance graph has 6,221,665 nodes and 11,210,114 edges, and the transfer graph 7,884,602 nodes and 25,001,216 edges. The solver graphs of Table 4.1 leave out 2,602 and 2,246 addresses whose pairs all carry a value weight of zero in floating point. On the same contracts the allowance graph has 44.8% as many edges as the transfer graph, because an owner–spender pair is an allowance edge only while its latest allowance of one of the five tokens is positive, whereas every pair of addresses that exchanged a non-zero transfer of these tokens in the window is a transfer edge. The sampling is the same for both, so the differences between the scores trace back to their edges and restarts.

The five tokens hold most of the extracted rows: of the ego-network events of the plan's cohort in the observation window, the revised analysis keeps 224,954,202 approvals (57.7%) and 1,149,168,538 transfers (74.7%), and every kept transfer has a price. Table 8.4 in Appendix 8.4.2 gives these figures per token.

4.1.2 Implementation

Every algorithm runs on one Python implementation of PageRank ($d = 0.85$, $\varepsilon = 10^{-8}$, $I_{\max} = 300$). EndorseRank and AWP differ in their edge lists and restart vectors, and C-PR mixes its two edge lists node by node (Equation (3.9)). Each evaluation run writes its scores, per-proxy statistics and bootstrap intervals to one summary file, and the benchmark its timings to another. The tables of this chapter come from runs made on 10 October 2026 from 11:36 UTC, after the revision had been committed. At $d = 0.85$ every solve converged well inside the iteration cap. Appendix 8.1 lists the runs, the repository revision, the configuration file and the summary files.

4.2 Computational benchmark (Claim C5)

Claim C5 concerns what each score costs to compute on the same input. Following the timing procedure of Section 3.11, Table 4.1 reports for each algorithm, on the full graphs of the matched cohort, the mean solve time and its standard deviation over five timed runs after one untimed warm-up, the peak memory of the warm-up solve, the number of iterations to convergence and the nodes and edges of the solver graph. Each graph and setting was timed once, so the same measurement reappears as the full-cohort row of Table 4.2 and as the $d = 0.85$ row of Table 4.5; Table 8.2 in Appendix 8.1.4 lists every timed run.

Table 4.1: Computational benchmark on the full graphs of the matched cohort ($n = 15{,}107$ contracts; AWP/EndorseRank runtime ratio 2.2× at an edge ratio of 2.2×). Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up, on which alone peak memory is traced; SD over the same 5 runs. A timed call turns an edge list into scores: node indexing, sparse-matrix assembly, row normalisation, restart vector and power iteration. $|V|$ and $|E|$ are the nodes and distinct edges of each method's solver graph. C-PR walks on the union of the allowance layer and the transfer layer, so its $|E|$ counts the distinct pairs of both; S-PR runs on the transfer layer, and its runtime and iterations include the solve of the allowance layer that supplies its restart vector. Table 8.2 lists every run.

Method	Runtime (s)	SD (s)	Peak mem. (MB)	Iters	$\lvert V\rvert$	$\lvert E\rvert$
EndorseRank	10.294	0.601	879.7	69	6,219,063	11,210,114
AWP	22.367	1.825	1,712.6	92	7,882,356	25,001,216
C-PR ($\lambda = 0.5$)	36.162	2.112	1,768.1	79	9,151,010	31,552,359
S-PR	34.422	3.833	1,235.8	158	7,884,602	25,018,892

EndorseRank solves the allowance graph in 10.29 s and AWP the transfer graph in 22.37 s, so AWP takes 2.17 times as long, at an edge ratio of 2.23. Each power iteration costs a sparse product over the $|E|$ edges and several vector operations over the $|V|$ nodes (Appendix 8.3.3). Divided by the iterations, assembly included, AWP's solve takes 0.243 s and EndorseRank's 0.149 s, 1.63 times as long, between the node ratio of 1.27 and the edge ratio, and AWP needs 1.33 times as many iterations (92 against 69); together the two factors give the time ratio. On the same graphs that ratio moves from 2.55 to 1.95 as d rises (Section 4.5.1), so its closeness to the edge ratio at $d = 0.85$ is

no general rule. Both scores use the same solver, so the difference in speed comes from the size of each graph and from the iterations its walk needs there, which the benchmark does not separate; it is no advantage of the EndorseRank algorithm.

C-PR walks on the union of both edge sets (31,552,359 distinct pairs, 9,151,010 nodes), but its solver keeps one matrix per layer and multiplies both in every iteration, up to about 36.2 million stored entries, 1.45 times AWP's edges. It converges in fewer iterations (79 against 92) and takes 36.16 s, 1.62 times as long as AWP, more than these counts predict. S-PR runs on the transfer layer and takes 34.42 s. Both its time and its iteration count include the solve of the allowance layer that supplies its restart vector: the 158 iterations are 63 for that solve plus 95 for the seeded transfer walk. Every algorithm that walks the transfer layer, alone or with allowance edges, pays for its size.

Two caveats attach to this result. The time covers only the solver call on edge lists built in advance; a full refresh would add parsing, decoding and edge-list construction, which all algorithms share wholly or in part. And a faster method is not necessarily a better one.

4.2.1 Runtime on subsamples of the matched cohort

Table 4.2 reports the runtimes of EndorseRank and AWP at each stage of Section 4.1.1, their ratio, the edge ratio and the nodes and edges of both solver graphs.

The stages of Section 3.12.1 change the edge ratio as well as the size: the transfer graph has 3.98 times as many edges as the allowance graph at $n = 1{,}000$, 2.49 times at $n = 2{,}000$, 2.60 at $n = 5{,}000$, 2.57 at $n = 10{,}000$ and 2.23 on the full cohort. If the time of a solve followed $|E|$ alone, the time ratio would follow these edge ratios; it is 2.89, 4.55, 3.60, 2.70 and 2.17. From $n = 10{,}000$ on, the time ratio stays within 0.13 of the edge ratio. At the three smaller stages, where EndorseRank's solve takes 0.58 to 2.50 s, it does not: it is 0.73 times the edge ratio at $n = 1{,}000$, although AWP needs 96 iterations and EndorseRank 56, 1.83 times at $n = 2{,}000$, where AWP needs 98 and EndorseRank 45, and 1.39 times at $n = 5{,}000$. The stages are subsamples of one cohort graph, so they show how runtime grows with the edges up to the full graph and not beyond

Table 4.2: Runtime of EndorseRank (ER) and AWP against the size of the evaluation set: deterministic SHA256 subsamples of the matched cohort (seed `contract-benchmark-v1`) up to the full cohort of $n = 15{,}107$ contracts. Each stage keeps the extracted edges incident to its sampled contracts, so $|V|$ and $|E|$ of both solver graphs are reported per stage. AWP/ER time and AWP/ER $|E|$ are the ratios of the two runtimes and of the two edge counts in the same row. Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up. [a]The full-cohort row is the measurement of Table 4.1, reused, not timed again.

n	ER (s)	AWP (s)	AWP/ER time	AWP/ER $\lvert E\rvert$	ER $\lvert V\rvert$	ER $\lvert E\rvert$	AWP $\lvert V\rvert$	AWP $\lvert E\rvert$
1,000	0.580	1.677	2.9×	4.0×	404,563	424,888	1,124,621	1,692,513
2,000	0.939	4.272	4.6×	2.5×	1,112,404	1,263,471	1,917,597	3,148,826
5,000	2.496	8.995	3.6×	2.6×	2,104,606	2,988,450	3,539,528	7,769,268
10,000	6.220	16.821	2.7×	2.6×	4,291,078	6,673,167	6,247,634	17,180,184
15,107[a]	10.294	22.367	2.2×	2.2×	6,219,063	11,210,114	7,882,356	25,001,216

it. Figure 4.1 plots both runtimes against the stage size, with the edge count of each graph beside its point.

4.3 Same-window alignment (Claim C4)

Claim C4 asks what each score agrees with in the window it is computed from. Table 4.3 reports the family-mean Kendall τ of every score on the transfer, allowance and Sybil-stability families, each with a 95% paired-bootstrap confidence interval. The raw degrees have no row because in this window they are proxies: in-degree belongs to the transfer family and in-approve degree to the allowance family. Apart from the secondary part of rule A, fixed in the analysis plan (Section 3.7) and applied here to the revised scores as a re-analysis, none of the comparisons below was fixed in advance (Chapter 3).

The scores divide by the edges they walk on. The two that walk on transfers alone, AWP and C-PR ($\lambda = 0$), reach 0.572 ($[0.564, 0.580]$) and 0.659 ($[0.653, 0.666]$) on the transfer family and 0.193 ($[0.182, 0.203]$) and 0.225 ($[0.214, 0.235]$) on the allowance family; S-PR, whose restarts come from the allowance walk, reaches 0.629 ($[0.621, 0.636]$) and 0.323 ($[0.313, 0.333]$). The two that walk on allowances alone, EndorseRank and C-PR ($\lambda = 1$), reach 0.585 ($[0.578, 0.592]$) and 0.597

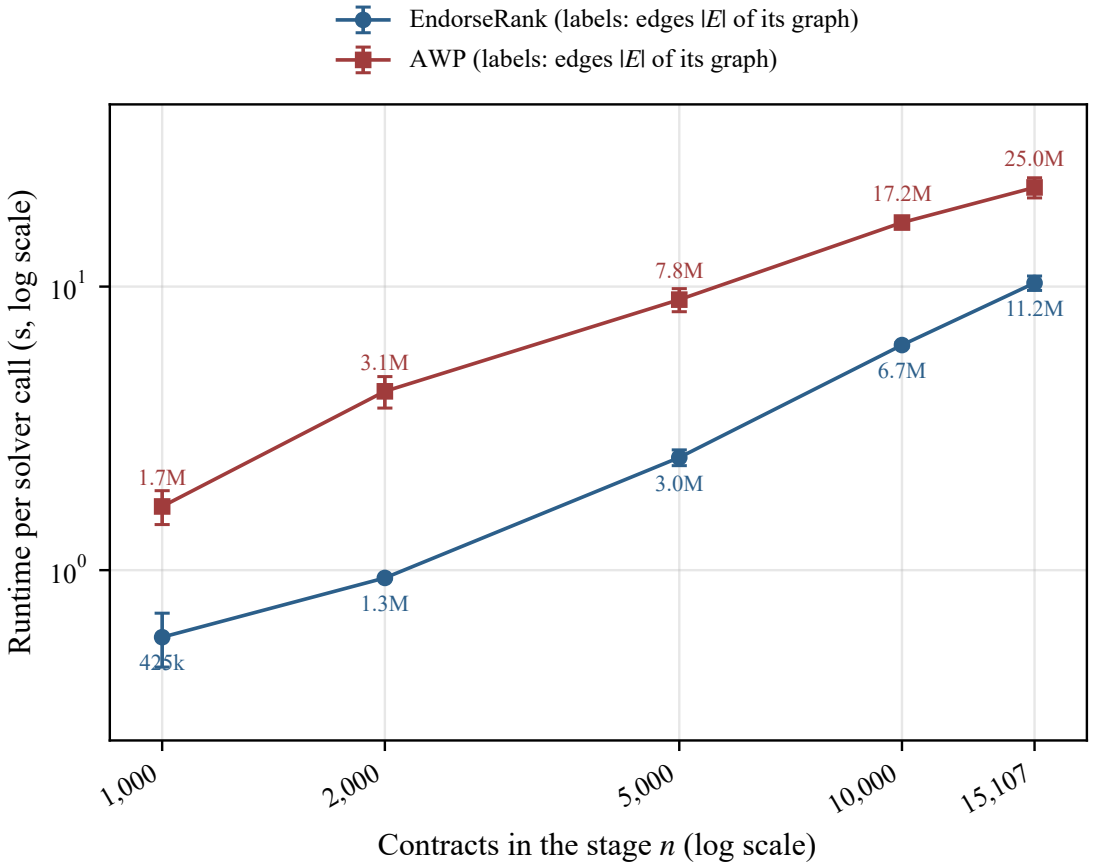


Figure 4.1: Wall-clock time of one solver call of EndorseRank and AWP against the number of contracts in each stage of Table 4.2, both axes on a log scale; the label beside each point gives the edge count $|E|$ of that graph, and error bars are the standard deviations over five timed runs.

Table 4.3: Same-window family-mean Kendall τ with bootstrap intervals for every score of this study (matched cohort, $n = 15{,}107$ contracts). The scores are EndorseRank and AWP, the coupled operator C-PR at five values of λ and the endorsement-seeded walk S-PR; $\lambda = 1$ is the allowance layer walked alone (latest allowances in USD, each capped) and $\lambda = 0$ the transfer layer walked alone (USD amounts discounted by the decay of AWP); both restart uniformly. The raw degrees have no row because they are proxies of the transfer and allowance families; EndorseRank with activity restarts is in Table 4.8. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Method	Transfer	Allowance	Sybil
EndorseRank	0.158 [0.147, 0.169]	0.585 [0.578, 0.592]	0.007 [-0.001, 0.015]
AWP	0.572 [0.564, 0.580]	0.193 [0.182, 0.203]	0.395 [0.387, 0.403]
C-PR ($\lambda = 1$)	0.162 [0.151, 0.173]	0.597 [0.590, 0.604]	0.011 [0.002, 0.019]
C-PR ($\lambda = 0$)	0.659 [0.653, 0.666]	0.225 [0.214, 0.235]	0.404 [0.396, 0.413]
C-PR ($\lambda = 0.25$)	0.385 [0.374, 0.395]	0.457 [0.449, 0.465]	0.155 [0.147, 0.164]
C-PR ($\lambda = 0.5$)	0.365 [0.355, 0.375]	0.477 [0.469, 0.485]	0.142 [0.134, 0.151]
C-PR ($\lambda = 0.75$)	0.344 [0.333, 0.354]	0.492 [0.484, 0.500]	0.129 [0.121, 0.138]
S-PR	0.629 [0.621, 0.636]	0.323 [0.313, 0.333]	0.398 [0.390, 0.406]

($[0.590, 0.604]$) on the allowance family and 0.158 ($[0.147, 0.169]$) and 0.162 ($[0.151, 0.173]$) on the transfer family, and the interior mixtures of C-PR lie between the two layers. On the Sybil-stability family the scores that walk on transfers lead, AWP with 0.395 ($[0.387, 0.403]$), while EndorseRank stays at 0.007 ($[-0.001, 0.015]$). Part of this agreement is built in. The main proxies of each family come from the same events as one of the graphs, in-degree and in-value from the transfers and in-approve degree and value from the latest allowances, and a PageRank smooths the in-degree of its own graph. Agreement with the proxies of a score's own edge type is therefore an intended-construct check and cannot count as an external test. The holdout of Section 4.6 asks the same question out of window.

Appendix 8.5.2 breaks the family means of EndorseRank and AWP into their proxies. AWP's largest coefficient is with transfer in-degree ($\tau = 0.578$, $[0.569, 0.587]$); with in-value it is almost as high (0.566, $[0.558, 0.575]$). That is what its weights lead one to expect: with the slope set at the median transfer (Section 3.6.1), an edge weight grows with the amounts transferred as well as with their number. In the registered analysis, with $b = 1$ on raw base units, V counted nearly

every transfer about once, and AWP followed the in-degree much more closely than the in-value (Appendix 8.6). It does not collapse to the in-degree either. Propagation, time weights and restarts at active senders shift a measurable share of pairwise orderings, and out of window AWP ranks new senders above the in-degree (Section 4.6). On the Sybil-stability family AWP follows tenure (0.508, $[0.499, 0.517]$) and active months (0.549, $[0.540, 0.558]$), and it is positive on the inbound-counterparty ratio (0.127, $[0.114, 0.141]$).

EndorseRank follows the in-approve degree (0.655, $[0.648, 0.662]$) more closely than the in-approve value (0.515, $[0.506, 0.524]$). Its value weight is 0.8 at twice the median allowance of 62 dollars, so an allowance of a few hundred dollars and an unlimited one weigh nearly alike, whereas the in-approve value counts each allowance up to the cap of 20,000 dollars. Because every cohort contract received approvals from at least three owners, EndorseRank leaves few ties, so its coefficients rest on an ordering of nearly every contract (Section 4.4). C-PR ($\lambda = 1$) walks the same allowance graph with capped amounts, also restarts uniformly, and stays within 0.02 of EndorseRank on every family.

The first half of Claim C4 is supported in a narrow sense. Every score that walks on one edge type agrees most with the proxies of that edge type, with an interval well clear of zero, and the mixtures of C-PR lie between their two layers on every family. The agreement is partly built in. Section 4.7.1 locates C-PR within the window.

4.4 Rank divergence (H1 / RQ1)

Hypothesis H1 and RQ1 concern whether EndorseRank and AWP put the same contracts in different orders. The inter-method Kendall τ_b between the two score vectors is 0.174, with interval $[0.163, 0.185]$ (last row of Table 4.4), so the interval excludes both zero and one. The two rankings are related, yet far from interchangeable. Of the $114{,}103{,}171$ contract pairs, 96.7% are ordered by both scores, and 41.2% of those pairs are discordant.

The marginal score distributions are in Figure 4.2, and ties play almost no part in them: EndorseR-

ank, with $13{,}625$ distinct scores, ties 0.17% of the contract pairs, and AWP, with $12{,}435$, 3.13%. Under each score the largest tie block is formed by the contracts that score zero: 535 under EndorseRank and $2{,}672$ under AWP, that is, the 523 and $2{,}670$ contracts outside each graph and 12 and 2 whose only pairs carry amounts too small for the value weight to differ from zero. Under EndorseRank 166 further contracts share the score $c = 7.64 \times 10^{-8}$ of a node without in-edges; they are in the allowance graph only as owners, and every other cohort contract has received a positive allowance and scores above c. A marginal distribution cannot show how two scores order the same contracts, so Figure 4.3 plots the two rankings against each other for a random sample of $3{,}000$ contracts. Contracts that both scores rank near the top, or both near the bottom, lie close to the identity line; in between, the points spread over most of the square.

The zero blocks come from one rule of this pipeline: a contract missing from a graph scores zero (Section 3.6). The analysis plan fixed the alternative, keeping the missing contracts as isolated nodes, as a sensitivity analysis. Under uniform teleportation an isolated node scores like any node without in-edges, so the 523 contracts outside the allowance graph receive c, the lowest score that any node of the graph can have, and join the 166 contracts at c near the bottom of the order. The 12 contracts whose only pairs carry zero weight are nodes of the graph that the solver leaves out, so they keep the score zero. No pair changes direction: the pairs between the 523 and the 166 become ties, and those between the 523 and the 12 become ordered. AWP's $2{,}670$ missing contracts were handled the same way; under AWP's restarts an isolated contract that has sent nothing receives no restart mass, so it scores zero either way. Table 4.4 gives the result. No coefficient changes at the third decimal, and the agreement between the two scores stays at 0.174 ($[0.163, 0.185]$). Because the cohort was defined by approvals received, nearly every cohort contract is a spender in the allowance graph, and the rule for missing contracts decides almost nothing.

H1 is supported. The two scores order the cohort differently, and the difference does not come from ties or from the rule for missing contracts. A protocol that relies on transfer-based scores cannot assume that its ordering of contracts survives a switch to allowance-based scores or a blend with them. This divergence, the second half of Claim C4, is also the premise of the coupled walk: each

Table 4.4: Same-window family-mean Kendall τ with contracts missing from a graph scored zero or kept as isolated nodes (matched cohort, $n = 15{,}107$ contracts). The main analysis scores a contract that is missing from a method's graph zero; keeping it as an isolated node is the sensitivity analysis fixed in the analysis plan. The rule concerns 523 contracts for EndorseRank and 2,670 for AWP (80 are missing from both graphs). Under EndorseRank's uniform restarts an isolated node gets the score of a node without in-edges. AWP restarts only at addresses that sent a transfer, so an isolated contract keeps the score zero and AWP's coefficients are the same under both rules. The last row is the agreement between the two scores. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Missing contracts scored zero		Missing contracts as isolated nodes	
Family	EndorseRank	AWP	EndorseRank	AWP
Transfer	0.158 [0.147, 0.169]	0.572 [0.564, 0.580]	0.158 [0.146, 0.169]	0.572 [0.564, 0.580]
Allowance	0.585 [0.578, 0.592]	0.193 [0.182, 0.203]	0.585 [0.579, 0.592]	0.193 [0.182, 0.203]
Sybil stability	0.007 [-0.001, 0.015]	0.395 [0.387, 0.403]	0.007 [-0.002, 0.015]	0.395 [0.387, 0.403]
EndorseRank vs. AWP	0.174 [0.163, 0.185]		0.174 [0.163, 0.185]	

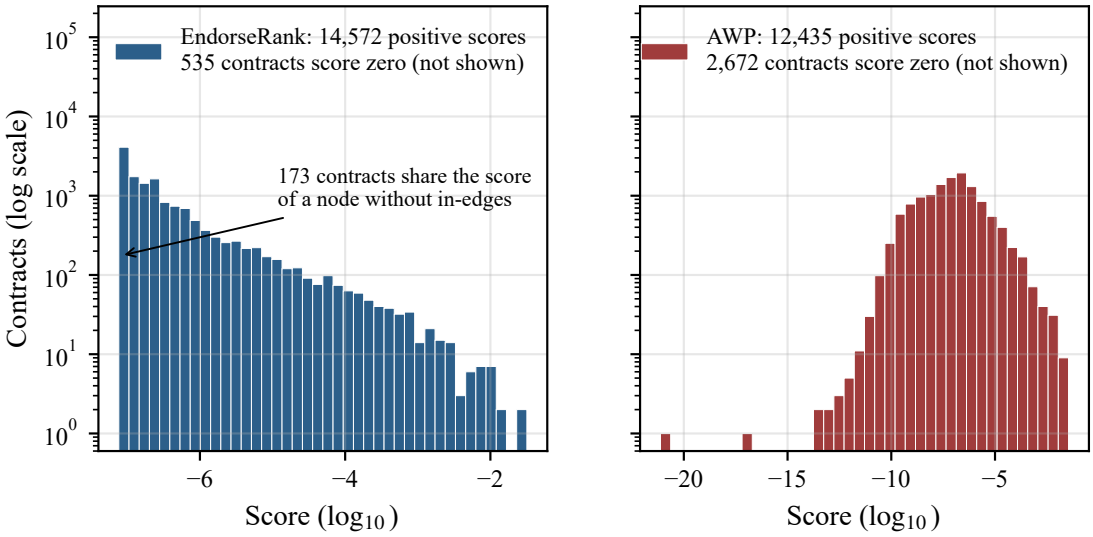


Figure 4.2: Marginal EndorseRank and AWP score distributions on the matched cohort ($n = 15{,}107$ contracts), drawn on a $\log_{10}$ scale for contracts with a positive score; counts are on a log scale. Zero-score contracts, which no edge of positive weight in the respective graph touches, are left out: 535 for EndorseRank and 2,672 for AWP.

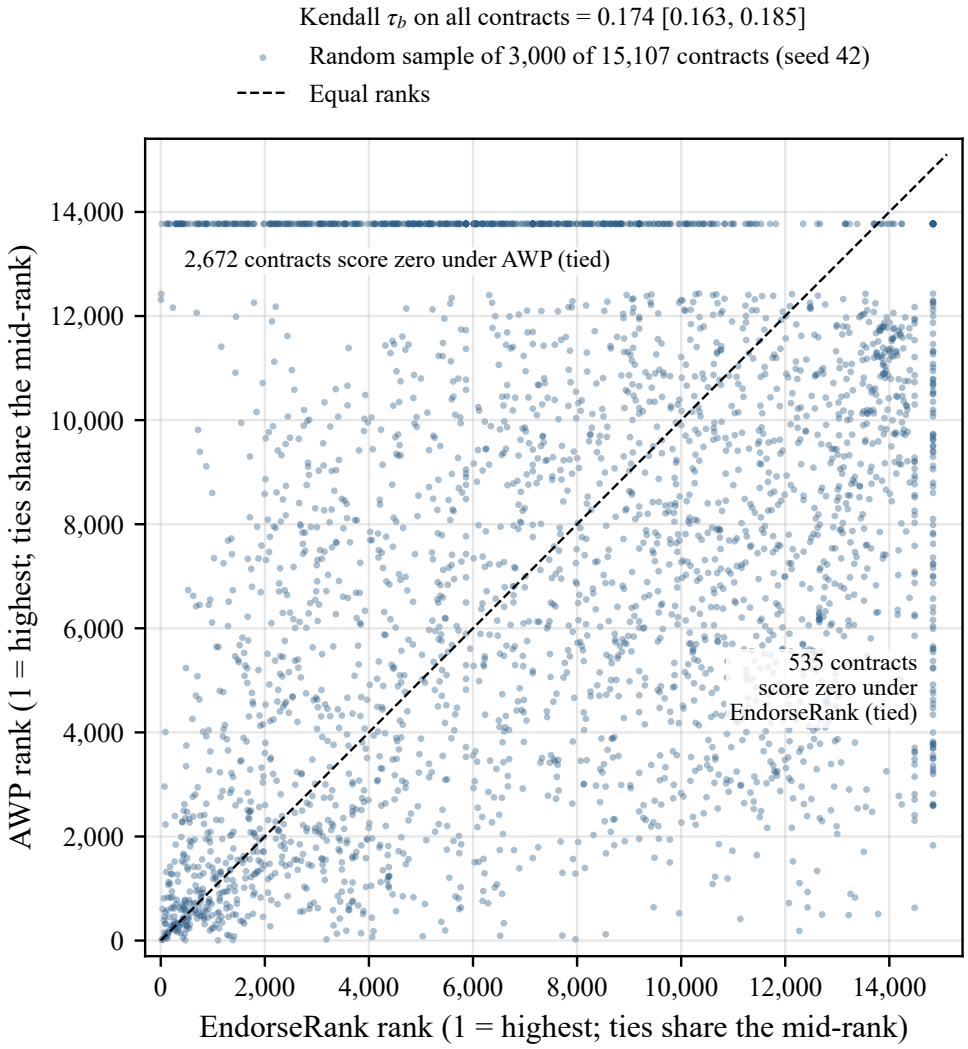


Figure 4.3: Rank–rank scatter of EndorseRank against AWP on the matched cohort (random sample of 3,000 of the 15,107 contracts, seed 42). Tied scores share their average rank, as in Kendall's τ_b. The column at the right edge holds the sampled contracts among the 535 that score zero under EndorseRank and the row at the top those among the 2,672 that score zero under AWP. A point on the dashed identity line has the same rank in both orderings; the legend gives τ_b on all contracts with its 95% interval.

single-layer walk orders the contracts by one kind of act, and Section 4.7 asks whether one walk over both edge types does better than either.

4.5 Robustness and sensitivity checks

Three checks, run for EndorseRank and AWP, ask whether the results hinge on one damping value, one slope of the value weight or one sample definition. The value-slope check takes the place of the plan's token check, which lost its point once every graph keeps five tokens (Section 3.12.3). The first two hold the sample fixed; the third varies the sample definition itself, so its coefficients are expected to move, and it is not evidence of robustness. A fourth check asks how much EndorseRank depends on its restart rule.

4.5.1 Damping

Table 4.5 recomputes the family means at $d \in \{0.75, 0.85, 0.95\}$ on the same graphs.

Table 4.5: Robustness: family-mean Kendall τ of EndorseRank (ER) and AWP under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on the matched cohort ($n = 15{,}107$ contracts); all. = allowance, tr. = transfer, Syb. = Sybil stability. Runtimes are means of five timed solver calls on the full graphs; the $d = 0.85$ runtimes are the measurements of Table 4.1, reused, not timed again (Table 8.2 lists every run). CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER Syb.	AWP Syb.	ER (s)	AWP (s)
0.75	0.586 [0.580, 0.594]	0.193 [0.182, 0.203]	0.158 [0.146, 0.168]	0.570 [0.562, 0.578]	0.007 [-0.002, 0.015]	0.397 [0.389, 0.406]	5.984	15.236
0.85	0.585 [0.578, 0.592]	0.193 [0.182, 0.203]	0.158 [0.147, 0.169]	0.572 [0.564, 0.580]	0.007 [-0.001, 0.015]	0.395 [0.387, 0.403]	10.294	22.367
0.95	0.583 [0.577, 0.591]	0.192 [0.181, 0.202]	0.158 [0.147, 0.169]	0.577 [0.569, 0.585]	0.008 [-0.001, 0.016]	0.391 [0.384, 0.400]	31.233	60.962

For EndorseRank no family mean moves by more than 0.003 across the three values. For AWP the transfer coefficient rises from 0.570 ($[0.562, 0.578]$) to 0.577 ($[0.569, 0.585]$) as d rises, and the Sybil-stability coefficient falls from 0.397 ($[0.389, 0.406]$) to 0.391 ($[0.384, 0.400]$). Neither score's pattern is an artifact of the conventional $d = 0.85$. EndorseRank's stability is largely structural. For a spender whose owners receive no approvals, EndorseRank equals $c\,(1 + d\,\delta(v))$, with $\delta(v)$ the

owner-normalized in-strength (Section 5.3.2), and the order that this formula gives among such spenders does not depend on d. At T_{obs} the formula holds within 1% for all 12,039 such spenders among the 14,406 cohort contracts with in-edges, so the rank correlations can hardly move. Runtime grows with the number of iterations, which rises with d: EndorseRank takes 5.98, 10.29 and 31.23 s (41, 69 and 217 iterations) and AWP 15.24, 22.37 and 60.96 s (55, 92 and 261), time ratios of 2.55, 2.17 and 1.95 at an edge ratio of 2.23. Figure 4.4 plots the family means.

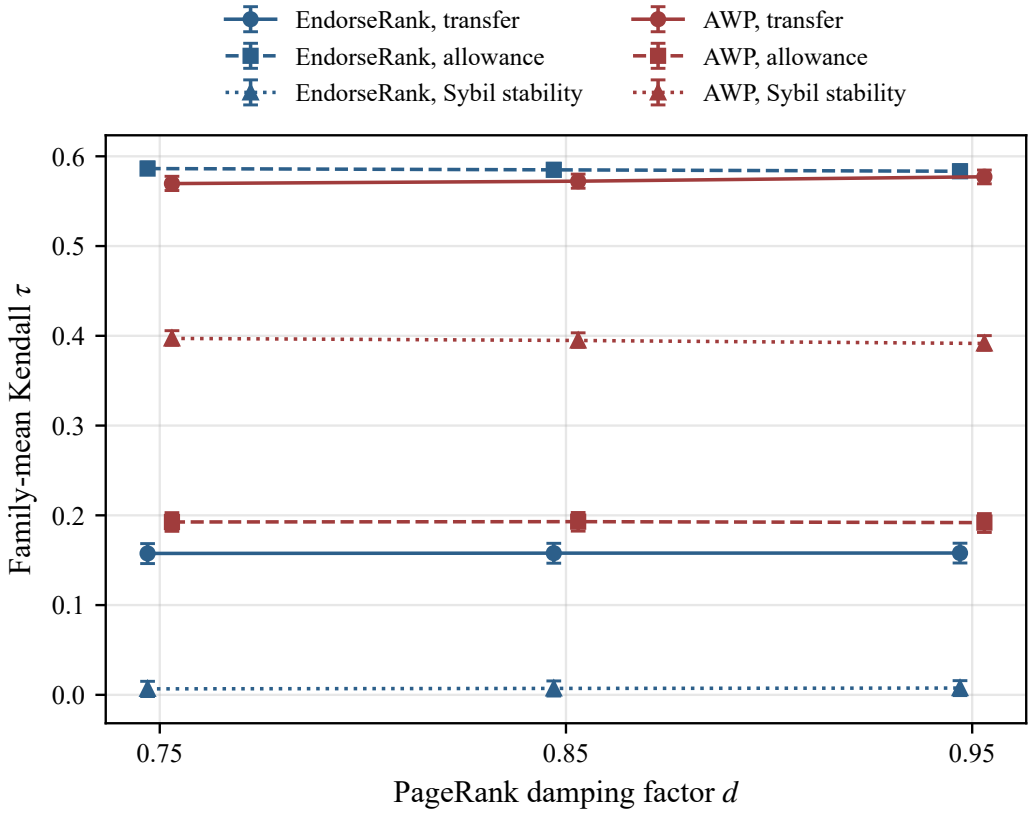


Figure 4.4: Damping sensitivity of the family-mean Kendall τ of EndorseRank and AWP on the transfer, allowance and Sybil-stability families across $d \in \{0.75, 0.85, 0.95\}$, with 95% bootstrap intervals (values from Table 4.5; matched cohort, $n = 15{,}107$ contracts).

4.5.2 Value-slope sensitivity

Table 4.6 recomputes EndorseRank and AWP at T_{obs} with the slope b of the value weight divided and multiplied by ten (Section 3.12.3), on the same graphs and proxies.

Table 4.6: Robustness: family-mean Kendall τ on the matched cohort ($n = 15{,}107$ contracts) when the slope b of the value weight V is divided or multiplied by ten, next to the main analysis. The main slopes put $V = 1/2$ at the median amount, \$200 for transfers and \$62 for allowances; $b/10$ moves that point to ten times the median and $10b$ to a tenth of it. The rank agreement τ of each variant with the main score of the same method (EndorseRank and AWP) is 0.973 and 0.973 at $b/10$, and 0.968 and 0.971 at $10b$. No intervals were computed for the variants.

Method	Transfer			Allowance			Sybil		
	$b/10$	b	$10b$	$b/10$	b	$10b$	$b/10$	b	$10b$
EndorseRank	0.157	0.158	0.159	0.588	0.585	0.580	0.006	0.007	0.009
AWP	0.577	0.572	0.567	0.193	0.193	0.191	0.397	0.395	0.393

Under either slope no family mean moves by more than 0.0054, and the pattern stays: EndorseRank leads on the allowance family, and AWP on the transfer and Sybil-stability families. The steeper slope $10b$, which puts the half weight at a tenth of the median amount and brings an edge weight closer to a count of events, lowers EndorseRank's allowance coefficient from 0.585 to 0.580 and AWP's transfer coefficient from 0.572 to 0.567; the flatter slope $b/10$ raises them to 0.588 and 0.577. Each variant orders the contracts almost as the main score of its method does (τ from 0.968 to 0.973). No intervals were computed for the variants.

4.5.3 Sample-definition sensitivity

Table 4.7 recomputes the family means on the subsamples of the runtime scaling.

EndorseRank moves most: its allowance coefficient is 0.715 at $n = 1{,}000$ and falls with every stage to 0.585 on the full cohort, and its transfer coefficient falls from 0.205 to 0.158. AWP stays within 0.018 of its full-cohort values on every family. No intervals were computed for the stages. None of these shifts is evidence about the scores. A stage keeps only the edges incident to its sampled contracts, so the paths that pass through unsampled cohort contracts are cut, and each score comes closer to a count of the sampled contract's own edges, which is what the proxies count. We therefore report the table as a sensitivity analysis of the sample definition (Section 3.12.4), and its values must not be compared with those of Table 4.3.

Table 4.7: Sensitivity of the family means to the evaluation set: family-mean Kendall τ of EndorseRank (ER) and AWP on deterministic SHA256 subsamples of the matched cohort (seed `contract-benchmark-v1`, full cohort $n = 15{,}107$ contracts), each stage scored on the edges incident to its contracts; all. = allowance, tr. = transfer, Syb. = Sybil stability. Rows are evaluated on different contract sets, so their τ values are not comparable with Table 4.3; they show how much the statistic moves with the evaluation population. No intervals were computed for the stages. Runtimes are the measurements of Table 4.2.

n	ER all.	AWP all.	ER tr.	AWP tr.	ER Syb.	AWP Syb.	ER (s)	AWP (s)	AWP/ER time
1,000	0.715	0.196	0.205	0.586	0.038	0.408	0.580	1.677	2.9×
2,000	0.686	0.199	0.196	0.588	0.033	0.411	0.939	4.272	4.6×
5,000	0.634	0.211	0.195	0.571	0.034	0.391	2.496	8.995	3.6×
10,000	0.607	0.196	0.167	0.571	0.013	0.394	6.220	16.821	2.7×
15,107	0.585	0.193	0.158	0.572	0.007	0.395	10.294	22.367	2.2×

4.5.4 Restarts of EndorseRank

EndorseRank restarts the walk uniformly, while AWP restarts it at addresses in proportion to their recent sending activity (Section 3.6.2). The analysis plan fixed uniform restarts for EndorseRank and named the variant as a score to report beside it, without fixing any comparison between the two. Table 4.8 scores both versions of EndorseRank, with AWP beside them.

Table 4.8: EndorseRank with uniform restarts and with restarts weighted by approving activity, AWP's rule applied to the allowance graph, next to AWP: family-mean Kendall τ on the matched cohort ($n = 15{,}107$ contracts) and τ with the two spender labels in W0 ($n = 17{,}050$ spender contracts, scores frozen at 31 March 2026) and W1 ($n = 18{,}152$, frozen at 30 June 2026). Both EndorseRank rows weight each latest allowance by $\sigma(\Delta t)V(z)$. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

		Same window, matched cohort			W0, spenders		W1, spenders	
Method	Restarts	Transfer	Allowance	Sybil	New approval pairs	New senders	New approval pairs	New senders
EndorseRank	uniform	0.158 [0.147, 0.169]	0.585 [0.578, 0.592]	0.007 [-0.001, 0.015]	0.231 [0.220, 0.242]	0.200 [0.189, 0.210]	0.234 [0.222, 0.244]	0.194 [0.184, 0.205]
EndorseRank	by approving activity	0.164 [0.153, 0.173]	0.426 [0.418, 0.435]	0.049 [0.039, 0.057]	0.308 [0.299, 0.319]	0.269 [0.260, 0.278]	0.295 [0.285, 0.304]	0.242 [0.232, 0.253]
AWP	by sending activity	0.572 [0.564, 0.580]	0.193 [0.182, 0.203]	0.395 [0.387, 0.403]	0.196 [0.181, 0.210]	0.346 [0.336, 0.356]	0.169 [0.157, 0.183]	0.330 [0.321, 0.340]

With AWP's restarts, EndorseRank's same-window allowance coefficient falls from 0.585 ($[0.578, 0.592]$) to 0.426 ($[0.418, 0.435]$), and its transfer coefficient changes little (0.164,

$[0.153, 0.173]$, against 0.158, $[0.147, 0.169]$). The restart mass then goes to the owners that approve most, so the score weights a spender by how active its owners are as well as by how many there are. Out of window the variant ranks both spender labels higher in both windows. In W0 it reaches 0.308 ($[0.299, 0.319]$) on new approval pairs and 0.269 ($[0.260, 0.278]$) on new senders, $+0.077$ ($[0.071, 0.084]$) and $+0.070$ ($[0.064, 0.076]$) above the uniform version, and in W1 0.295 ($[0.285, 0.304]$) and 0.242 ($[0.232, 0.253]$), $+0.061$ ($[0.054, 0.067]$) and $+0.048$ ($[0.042, 0.053]$) above it. Table 4.8 gives the coefficients and Table 4.13 the W1 contrasts. These contrasts were computed after the labels were known. The uniform rule keeps the allowance construct in the same window, and the activity rule ranks later arrivals better: in W0 the variant ranks new approval pairs above the in-approve degree ($+0.059$, $[0.050, 0.067]$), where the uniform version falls below it (Section 4.6). Section 4.7.4 sets the variant beside C-PR.

4.6 Temporal holdout of the single-layer walks (RQ3)

A same-window check cannot separate a score from its own construct. The holdout instead tests whether centrality measured up to a cut-off predicts behavior after it. Following Section 3.7, scores on the W0 spender cohort are frozen at t_1 (31 March 2026), and two labels are then counted for each contract between 1 April and 30 June 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). Table 4.9 sets every score computed at t_1, the two raw-degree baselines included, against both labels. Table 4.10 gives the paired contrasts, first the two of rule A, fixed in the analysis plan before any log of the study was extracted, and then those computed after the labels were known. This section reads the single-layer walks and the raw degrees; rule A and the rows of C-PR and S-PR are taken up in Section 4.7.

The first finding answers the second part of RQ3 for the single-layer walks: whether a walk improves on its raw degree appears to depend on its restart rule, not on its edge type. AWP, which restarts at active senders, ranks new transfer senders at $\tau = 0.346$ ($[0.336, 0.356]$), above the transfer in-degree at t_1 (0.303, $[0.292, 0.315]$), a paired difference of $+0.043$ (interval

Table 4.9: Temporal holdout W0 on the spender cohort ($n = 17{,}050$ contracts): Kendall τ between scores frozen at 31 March 2026 and two labels counted on the contract between 1 April 2026 and 30 June 2026: the number of new owner–spender approval pairs and the number of addresses that sent a transfer to the contract for the first time. The cohort keeps the 17,050 contracts among the 17,167 spenders with a positive latest allowance at the freeze; 104 EOAs without code and 13 EOAs with an EIP-7702 delegation are left out. 2,070 contracts received at least one new pair and 1,695 at least one new sender. Baseline rows are the raw degrees at t_1 that each single-layer PageRank smooths. C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$) are the allowance and transfer layers of the coupled operator walked alone. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at t_1 (baseline)	0.250 [0.237, 0.262]	0.180 [0.169, 0.192]
Transfer in-degree at t_1 (baseline)	0.135 [0.121, 0.150]	0.303 [0.292, 0.315]
EndorseRank	0.231 [0.220, 0.242]	0.200 [0.189, 0.210]
AWP	0.196 [0.181, 0.210]	0.346 [0.336, 0.356]
C-PR ($\lambda = 1$)	0.223 [0.213, 0.233]	0.188 [0.177, 0.199]
C-PR ($\lambda = 0$)	0.160 [0.144, 0.174]	0.296 [0.285, 0.307]
C-PR ($\lambda = 0.25$)	0.253 [0.243, 0.264]	0.269 [0.257, 0.280]
C-PR ($\lambda = 0.5$)	0.252 [0.241, 0.262]	0.262 [0.250, 0.273]
C-PR ($\lambda = 0.75$)	0.249 [0.238, 0.260]	0.253 [0.242, 0.265]
S-PR	0.172 [0.157, 0.185]	0.295 [0.284, 0.305]

Table 4.10: Paired bootstrap differences on the spender holdout W0 ($n = 17{,}050$ contracts; 400 resamples shared by a and b, seed 42). The first block holds the two contrasts of the primary part of rule A, fixed in the analysis plan committed on 9 October 2026 before the study extracted any data and applied here to the revised scores (C-PR against C-PR ($\lambda = 1$) on new approval pairs and against C-PR ($\lambda = 0$) on new transfer senders). The second block was computed after the labels were known: the increment of C-PR ($\lambda = 1$), the allowance layer walked alone, over its raw degree at t_1, the same comparisons for S-PR, and the increments of EndorseRank and AWP over their raw degrees; C-PR's other contrasts are in Table 4.18. Primary part of rule A (C-PR no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new transfer senders): no worse holds, better fails, so the rule is not met. The differences between EndorseRank and AWP are in Table 8.10.

Contrast (a minus b)	Label	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Rule A, primary part, fixed in the plan						
C-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.252	0.223	+0.029	[0.022, 0.036]	1.000
C-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.262	0.296	-0.034	[-0.040, -0.029]	0.000
Computed after the labels were known						
C-PR ($\lambda = 1$) minus t_1 in-approve degree	new approval pairs	0.223	0.250	-0.027	[-0.034, -0.019]	0.000
S-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.172	0.223	-0.051	[-0.066, -0.038]	0.000
S-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.295	0.296	-0.001	[-0.004, 0.002]	0.265
EndorseRank minus t_1 in-approve degree	new approval pairs	0.231	0.250	-0.019	[-0.026, -0.011]	0.000
AWP minus t_1 in-degree	new transfer senders	0.346	0.303	+0.043	[0.038, 0.048]	1.000

$[0.038, 0.048]$). The transfer layer walked with uniform restarts does not: C-PR ($\lambda = 0$) reaches 0.296 ($[0.285, 0.307]$), slightly below the in-degree (no paired contrast was computed). On the allowance side, with uniform restarts, the count wins. In-approve degree at t_1 predicts new approval pairs with $\tau = 0.250$ ($[0.237, 0.262]$). EndorseRank reaches 0.231 ($[0.220, 0.242]$), a paired difference of -0.019 ($[-0.026, -0.011]$), and C-PR ($\lambda = 1$) 0.223 ($[0.213, 0.233]$), a difference of -0.027 ($[-0.034, -0.019]$); with activity restarts EndorseRank beats the count (Section 4.5.4). Within a three-month window, a spender gains more new owners the more it already has, as preferential attachment would predict (Barabási & Albert, 1999), and the local count captures this directly. In EndorseRank an owner's endorsement is shared among all the spenders it approves (Section 5.3.2), so a spender whose owners approve many other spenders ranks lower than its count alone would place it; on this label that sharing costs a little. These degree contrasts were computed after the labels were known.

Read across the whole spender cohort, each label is ranked best by scores that use its own edge type. On new approval pairs the C-PR mixtures (0.249, $[0.238, 0.260]$, to 0.253, $[0.243, 0.264]$) and the in-approve degree lead, followed by EndorseRank and C-PR ($\lambda = 1$), while AWP (0.196, $[0.181, 0.210]$), S-PR, C-PR ($\lambda = 0$) and the transfer in-degree (0.135, $[0.121, 0.150]$) stay lower. On new transfer senders AWP and the transfer in-degree lead, and the allowance-side scores stay lower and close together: EndorseRank reaches 0.200 ($[0.189, 0.210]$), C-PR ($\lambda = 1$) 0.188 ($[0.177, 0.199]$) and the in-approve degree 0.180 ($[0.169, 0.192]$), with overlapping intervals. EndorseRank gives $16{,}208$ distinct scores to the $17{,}050$ spenders, and its largest tie block holds 156.

Coverage does not explain the degree comparisons, but it explains the comparison across edge types on new approval pairs. Of the 17,050 spenders, 2,891, about one in six, have no transfer edge up to the freeze; AWP, C-PR ($\lambda = 0$) and S-PR score them zero, and only allowance edges order them. They gained new approval pairs slightly more often than the rest (13.3% against 11.9%) and new senders almost never (0.6% against 11.9%), so AWP's zeros lower its coefficient on the first label and raise it on the second. Table 4.11 restricts the holdout to the 14,094 spenders that had

received a transfer from another address before the freeze. There AWP still ranks new senders above the transfer in-degree (0.357, $[0.347, 0.367]$, against 0.301, $[0.289, 0.313]$; $+0.056$, $[0.049, 0.064]$), and EndorseRank stays below the in-approve degree on new approval pairs (0.236, $[0.224, 0.249]$, against 0.249, $[0.237, 0.264]$; -0.013, $[-0.020, -0.006]$). On new approval pairs AWP now reaches 0.298 ($[0.287, 0.308]$), above both (no paired contrast was computed), so the lead of the allowance-side scores on this label rests on the 2,956 spenders left out, which include the 2,891 that AWP scores zero. The restriction was chosen after the results were known. The row of C-PR in that table is read in Section 4.7.4.

Table 4.11: Spender holdout W0 restricted to the 14,094 of the 17,050 spender contracts that had received a transfer from another address before the freeze ($n = 14{,}094$). The other 2,956 have no inbound transfer edge from another address, and 2,891 spenders score zero under AWP. The restriction was chosen after the results were known. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42). [a]Computed afterwards from the saved scores, with the same resamples; its contrasts with AWP and the in-approve degree on this subset are in Table 4.18.

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
EndorseRank	0.236 [0.224, 0.249]	0.231 [0.219, 0.244]
AWP	0.298 [0.287, 0.308]	0.357 [0.347, 0.367]
C-PR ($\lambda = 0.5$)[a]	0.261 [0.250, 0.272]	0.284 [0.272, 0.296]
In-approve degree at t_1	0.249 [0.237, 0.264]	0.205 [0.193, 0.218]
Transfer in-degree at t_1	0.211 [0.196, 0.224]	0.301 [0.289, 0.313]
EndorseRank minus t_1 in-approve degree (new approval pairs)	$\Delta\tau = -0.013$ [-0.020, -0.006]	
AWP minus t_1 in-degree (new transfer senders)	$\Delta\tau = +0.056$ [0.049, 0.064]	

Two caveats apply to the whole table. Each label is the same construction carried into the future (new approvals, new senders), so none of these spender results bears on credit or loss; the trader labels below are built from neither edge type but are not credit labels either. Both labels are also sparse. Of the 17,050 spenders, 2,070 gained a new approval pair and 1,695 a new sender; every coefficient rests largely on how these subsets are ordered.

Other candidate labels were computed on the holdout, and they are reported here without being offered as support. Revocation counts, which for some tokens include allowances spent in full (Section 3.7), and the drain heuristic both rise with EndorseRank ($\tau = 0.337$, interval $[0.327, 0.347]$, and 0.234, interval $[0.224, 0.245]$) and with every other score, and revocations rise most steeply with

the raw in-approve degree (0.388, $[0.376, 0.398]$). For an endorsement reading the sign is wrong, since a trusted spender should lose approvals less often, not more. A spender with many owners sees more of everything owners do, revocations and drains included, so these labels cannot separate trust from use.

4.6.1 Trader outcomes (exploratory)

The trader run of Section 3.7.1 applies the W0 freeze to the GMX V2 contract traders and is reported as exploratory; its liquidation labels are left for the comparison of Section 4.8. Table 4.12 gives every score on the four profit labels of the 88 contract traders with at least three closes from April to June.

The walks that use transfer edges, the mixtures included, rank the number of profitable closes and the realized gain with intervals above zero: AWP reaches 0.155 ($[0.005, 0.297]$) and 0.196 ($[0.048, 0.351]$), and C-PR ($\lambda = 0$), the three mixtures and S-PR 0.150 ($[0.011, 0.272]$) to 0.166 ($[0.023, 0.290]$) and 0.270 ($[0.127, 0.398]$) to 0.301 ($[0.147, 0.436]$). The intervals of EndorseRank, C-PR ($\lambda = 1$) and both raw degrees contain zero, and the transfer in-degree is near zero on both labels (0.008, $[-0.135, 0.169]$, and 0.008, $[-0.149, 0.175]$). The paired contrasts separate the two sides only on realized gain, for the C-PR layers (-0.248, $[-0.433, -0.067]$), and not on profitable closes (EndorseRank minus AWP -0.005, $[-0.163, 0.155]$; Table 8.11). Both labels grow with how much a contract trades. On the two shares, which do not, every coefficient is negative. On the non-loss share every interval lies below zero, from -0.178 ($[-0.298, -0.042]$) for EndorseRank to -0.278 ($[-0.431, -0.119]$) for the in-approve degree; on the profitable share only that of the transfer in-degree does (-0.191, $[-0.329, -0.041]$). Part of this is built into the non-loss share (Section 4.8). None of the scores ranks how well a contract trades. The registered window gives C-PR a trader label of its own.

Table 4.12: W0 trader run: Kendall τ between scores frozen at 31 March 2026 and GMX V2 labels counted from 1 April 2026 to 30 June 2026, on the $n = 88$ GMX V2 contract accounts in the freeze-date graph with at least three closes in that window, 40 of which had received an approval by the freeze. The run is exploratory; the liquidation labels of W0 were left for the comparison (Table 4.19). Profitable closes and realized gain grow with the number of closes; the two shares do not. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at t_1 (baseline)	0.083 [-0.103, 0.238]	0.024 [-0.154, 0.202]	-0.146 [-0.317, 0.029]	-0.278 [-0.431, -0.119]
Transfer in-degree at t_1 (baseline)	0.008 [-0.135, 0.169]	0.008 [-0.149, 0.175]	-0.191 [-0.329, -0.041]	-0.197 [-0.350, -0.035]
EndorseRank	0.150 [-0.019, 0.295]	0.078 [-0.094, 0.226]	-0.055 [-0.196, 0.084]	-0.178 [-0.298, -0.042]
AWP	0.155 [0.005, 0.297]	0.196 [0.048, 0.351]	-0.124 [-0.265, 0.015]	-0.254 [-0.404, -0.104]
C-PR ($\lambda = 1$)	0.083 [-0.072, 0.223]	0.040 [-0.133, 0.190]	-0.074 [-0.231, 0.068]	-0.188 [-0.322, -0.047]
C-PR ($\lambda = 0$)	0.150 [0.011, 0.272]	0.288 [0.128, 0.438]	-0.101 [-0.243, 0.045]	-0.225 [-0.359, -0.083]
C-PR ($\lambda = 0.25$)	0.152 [0.015, 0.274]	0.294 [0.143, 0.424]	-0.084 [-0.233, 0.069]	-0.210 [-0.352, -0.061]
C-PR ($\lambda = 0.5$)	0.160 [0.030, 0.284]	0.283 [0.142, 0.410]	-0.084 [-0.227, 0.058]	-0.208 [-0.346, -0.061]
C-PR ($\lambda = 0.75$)	0.155 [0.025, 0.279]	0.270 [0.127, 0.398]	-0.082 [-0.221, 0.063]	-0.203 [-0.341, -0.059]
S-PR	0.166 [0.023, 0.290]	0.301 [0.147, 0.436]	-0.075 [-0.221, 0.072]	-0.221 [-0.365, -0.078]

4.6.2 The single-layer walks in the registered window W1

The registered window W1 (Section 3.7.2) freezes the scores at T_{obs}, 30 June 2026, and counts the same two spender labels from July to September 2026. Its registration and rule B concern C-PR and are reported in Section 4.7.3. At the freeze, 18,152 contract spenders held a positive latest allowance; 1,616 of them gained a new approval pair in the label window and 1,533 a new sender. Table 4.13 adds EndorseRank and its variant with activity restarts, which the registration does not name. None of their rows enters a rule, and the table's contrasts were computed after the labels were known, those of the variant from the saved W1 scores.

The W0 pattern holds for both single-layer walks in this window too. The in-approve degree at the freeze again ranks new approval pairs above EndorseRank (0.268, $[0.257, 0.279]$, against 0.234, $[0.222, 0.244]$; -0.034, $[-0.038, -0.029]$), and AWP again ranks new senders above the transfer in-degree (0.330, $[0.321, 0.340]$, against 0.308, $[0.299, 0.319]$; $+0.022$, $[0.017, 0.027]$). With activity restarts EndorseRank ranks new approval pairs above the degree ($+0.027$, $[0.021, 0.034]$), as it did in W0. On the liquidation-free close rate of the W1 traders EndorseRank reaches 0.297 ($[0.174, 0.418]$).

4.7 The coupled walk C-PR (RQ4, RQ5)

C-PR is the method of this thesis: one random walk over the US-dollar allowance layer and the US-dollar transfer layer, which at each node mixes the two transition rows with weight λ (Section 3.6.3). The analysis plan fixed one decision rule, and its object is C-PR: rule A asks the coupled walk to beat each of its layers walked alone (RQ4). The registration of W1 added rule B, which asks it to improve on its transfer layer on new data (RQ5). Both rules compare C-PR with its own layers, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$), because only that contrast changes the coupling and nothing else; C-PR also differs from EndorseRank and AWP in its amount weights and its uniform restarts. Both rules are applied to the revised scores as a re-analysis. This section places C-PR within the window, then reports rule A in W0 and rule B in W1, and ends with its contrasts with every other

Table 4.13: EndorseRank and its variant with activity restarts on the cohorts and labels of the window W1 (spenders $n = 18{,}152$ contracts, traders $n = 70$). Neither these rows nor the contrasts below them are part of rule B, and the contrasts were not named in the registration. The contrasts of EndorseRank with activity restarts were computed afterwards from the saved W1 scores. The AWP, C-PR and degree rows repeat Table 4.15, and every contrast uses the same contract resamples; the differences between EndorseRank and AWP are in Table 8.10, and those of C-PR in Table 4.18. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Spenders ($n = 18{,}152$)		Traders ($n = 70$)
Score at 30 June 2026	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.268 [0.257, 0.279]	0.205 [0.193, 0.215]	0.394 [0.261, 0.524]
Transfer in-degree at the freeze (baseline)	0.131 [0.119, 0.143]	0.308 [0.299, 0.319]	0.310 [0.185, 0.416]
EndorseRank	0.234 [0.222, 0.244]	0.194 [0.184, 0.205]	0.297 [0.174, 0.418]
EndorseRank, restarts by approving activity	0.295 [0.285, 0.304]	0.242 [0.232, 0.253]	0.288 [0.161, 0.405]
AWP	0.169 [0.157, 0.183]	0.330 [0.321, 0.340]	0.329 [0.215, 0.439]
C-PR ($\lambda = 0.5$)	0.250 [0.240, 0.260]	0.263 [0.254, 0.273]	0.350 [0.236, 0.461]
EndorseRank minus in-approve degree, new approval pairs	$\Delta\tau = -0.034$ [-0.038, -0.029]		
AWP minus transfer in-degree, new transfer senders	$\Delta\tau = +0.022$ [0.017, 0.027]		
EndorseRank (activity restarts) minus in-approve degree, new approval pairs	$\Delta\tau = +0.027$ [0.021, 0.034]		
EndorseRank (activity restarts) minus EndorseRank, new approval pairs	$\Delta\tau = +0.061$ [0.054, 0.067]		
EndorseRank (activity restarts) minus EndorseRank, new transfer senders	$\Delta\tau = +0.048$ [0.042, 0.053]		

score of the study, which decide nothing.

4.7.1 The coupled walk within the window

Two hybrids were defined in Section 3.6.3. C-PR mixes the US-dollar allowance and transfer layers in each node's transition row with weight λ, and its ablation S-PR keeps the transfer layer but restarts from the allowance walk. Table 4.14 gives their paired contrasts in this window: first the eight of Section 3.10 against the two layers walked alone, then six against EndorseRank and AWP, two of them computed afterwards from the saved scores. RQ4 and RQ5 are decided out of window. Here we only locate the hybrids on the checks that the single-layer scores were built to pass.

C-PR lies between its two layers on every family and stays nearer to its allowance layer. At the primary setting it reaches 0.477 ($[0.469, 0.485]$) on the allowance family, 0.120 below C-PR ($\lambda = 1$) ($[-0.127, -0.113]$), and 0.365 ($[0.355, 0.375]$) on the transfer family, 0.294 below C-PR ($\lambda = 0$) ($[-0.305, -0.284]$). As λ rises from 0.25 to 0.75 it moves toward the allowance layer on all three

Table 4.14: Same-window paired contrasts for the coupled operator (C-PR, $\lambda = 0.5$) and the endorsement-seeded walk (S-PR) against the two layers they are built from, the allowance layer (C-PR ($\lambda = 1$)) and the transfer layer (C-PR ($\lambda = 0$)), and of C-PR against EndorseRank and AWP (matched cohort, $n = 15{,}107$ contracts). Family-mean τ; the same contract resamples are used for a and b. The contrasts with the layers are a stricter check beside the secondary part of rule A, which compares C-PR's point estimate with the better layer's interval in Table 4.3. The secondary part of rule A, fixed in the analysis plan and applied here to the revised scores, is not met: C-PR's point estimate lies outside the interval of C-PR ($\lambda = 0$) on the transfer family and outside the interval of C-PR ($\lambda = 1$) on the allowance family, and C-PR does not exceed both layers on the Sybil-stability family. An interval below 0 means a is below b beyond sampling variation; one above 0 means it exceeds it. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42). [a]Computed afterwards from the saved scores, with the same resamples: C-PR against EndorseRank on the transfer family and against AWP on the allowance family, the family that each single-layer walk follows least.

Contrast (a minus b)	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Against the two layers walked alone					
C-PR transfer minus C-PR ($\lambda = 0$) transfer	0.365	0.659	-0.294	[-0.305, -0.284]	0.000
C-PR allowance minus C-PR ($\lambda = 1$) allowance	0.477	0.597	-0.120	[-0.127, -0.113]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.142	0.404	-0.262	[-0.273, -0.253]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.142	0.011	+0.131	[0.126, 0.137]	1.000
S-PR transfer minus C-PR ($\lambda = 0$) transfer	0.629	0.659	-0.031	[-0.035, -0.028]	0.000
S-PR allowance minus C-PR ($\lambda = 1$) allowance	0.323	0.597	-0.274	[-0.285, -0.263]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.398	0.404	-0.007	[-0.009, -0.004]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.398	0.011	+0.387	[0.376, 0.399]	1.000
Against EndorseRank and AWP					
C-PR transfer minus AWP transfer	0.365	0.572	-0.207	[-0.220, -0.194]	0.000
C-PR allowance minus EndorseRank allowance	0.477	0.585	-0.108	[-0.115, -0.101]	0.000
C-PR sybil-stability minus AWP sybil-stability	0.142	0.395	-0.253	[-0.263, -0.242]	0.000
C-PR sybil-stability minus EndorseRank sybil-stability	0.142	0.007	+0.135	[0.129, 0.141]	1.000
C-PR transfer minus EndorseRank transfer[a]	0.365	0.158	+0.207	[0.200, 0.215]	1.000
C-PR allowance minus AWP allowance[a]	0.477	0.193	+0.284	[0.274, 0.294]	1.000

families. AWP ranks the transfer family better than C-PR (-0.207, $[-0.220, -0.194]$), and EndorseRank the allowance family (-0.108, $[-0.115, -0.101]$). On the family that each single-layer walk follows least, C-PR leads that walk: it ranks the transfer family 0.207 above EndorseRank ($[0.200, 0.215]$) and the allowance family 0.284 above AWP ($[0.274, 0.294]$). A node with out-edges in only one layer follows that layer whatever λ is (Equation (3.9)); λ acts only at the nodes with out-edges in both layers. Almost every node of the allowance layer is an owner with out-edges, so the allowance rows enter the walk wherever an owner also sent a transfer that the extraction kept. On the Sybil-stability family C-PR exceeds its allowance layer ($+0.131$, $[0.126, 0.137]$) and lies below its transfer layer (-0.262, $[-0.273, -0.253]$).

On every family, S-PR falls below the layer that leads that family. Restarting the transfer walk from the allowance walk drops the transfer coefficient to 0.629 ($[0.621, 0.636]$; $\Delta\tau = -0.031$, $[-0.035, -0.028]$, against C-PR ($\lambda = 0$)), raises the allowance coefficient only as far as 0.323 ($[0.313, 0.333]$; $\Delta\tau = -0.274$, $[-0.285, -0.263]$, against C-PR ($\lambda = 1$)), and leaves Sybil-adjusted stability slightly below the transfer layer (-0.007, $[-0.009, -0.004]$). A restart vector changes where the walk lands after each teleport. It carries part of the allowance signal into the transfer walk, but less than walking the allowance edges does. This is the same-window counterpart of the finding that Section 4.7.2 reaches out of window.

As fixed in the analysis plan, the secondary part of rule A asks C-PR's point estimate to lie within the 95% interval of the better layer on the transfer and allowance families, and C-PR to exceed both layers on Sybil-adjusted stability. At $\lambda = 0.5$, C-PR fails all three parts, as in the registered analysis (Appendix 8.6). Its transfer coefficient (0.365) lies below the interval of C-PR ($\lambda = 0$) ($[0.653, 0.666]$), its allowance coefficient (0.477) lies below that of C-PR ($\lambda = 1$) ($[0.590, 0.604]$), and on Sybil-adjusted stability it exceeds only the allowance layer. The paired contrasts of Table 4.14, reported alongside the rule as a stricter check, point the same way. In the same-window tables the mixture gives up part of each layer's agreement with its own family and exceeds the better layer on no family. The holdout takes up the remaining question of whether the mixture has any predictive content that the single layers lack.

4.7.2 Rule A in the window W0

The coupled walk fails rule A out of window, although in an exploratory comparison it improves on its transfer layer walked alone. That rule, applied here to the revised scores as a re-analysis, asked C-PR to do no worse than C-PR ($\lambda = 1$) on new approval pairs and to beat C-PR ($\lambda = 0$) on new transfer senders in W0 (first block of Table 4.10). At $\lambda = 0.5$, C-PR lies above the allowance walk on new approval pairs ($\Delta\tau = +0.029$, interval $[0.022, 0.036]$), so the first condition holds, and below the transfer walk on new transfer senders ($\Delta\tau = -0.034$, interval $[-0.040, -0.029]$), so the second fails, and with it the primary part, as in the registered analysis (Table 8.12). Against the transfer walk alone, a comparison the rule did not include, C-PR does better. On new approval pairs C-PR reaches 0.252 ($[0.241, 0.262]$) against 0.160 ($[0.144, 0.174]$), a paired difference of $+0.092$ (interval $[0.080, 0.105]$, on the same resamples). The transfer walk, like AWP, scores zero the spenders without a transfer edge, and C-PR was not contrasted with it on the other spenders, so part of this gain may come from ordering those spenders. This contrast was computed after the labels were known; in the registered analysis the same contrast became the hypothesis F1 of the registration of W1 (Section 4.7.3). Across the three values of λ, C-PR is flat on new approval pairs (0.253, $[0.243, 0.264]$, at $\lambda = 0.25$; 0.249, $[0.238, 0.260]$, at $\lambda = 0.75$), and its point estimate on new transfer senders falls as λ rises, from 0.269 ($[0.257, 0.280]$) to 0.253 ($[0.242, 0.265]$) (no paired contrast was computed, and λ was not tuned).

The seeded ablation shows where the allowance signal has to sit. S-PR matches the transfer walk on the flow label ($\Delta\tau = -0.001$, interval $[-0.004, 0.002]$) and falls below the allowance walk on the endorsement label ($\Delta\tau = -0.051$, $[-0.066, -0.038]$), ranking new approval pairs only a little above the transfer walk (0.172, $[0.157, 0.185]$, against 0.160). When transfer edges decide where the walk goes, a restart distribution derived from endorsements changes little, although, like the transfer walk, S-PR scores zero the spenders without a transfer edge, so part of this gap may come from coverage. On the spender holdout, the allowance signal survives when allowance edges carry the walk, fully in the allowance walk and mixed in C-PR, and most of it is lost when they only

decide where the walk restarts.

4.7.3 Rule B in the registered window W1

The registration of Section 3.7.2 was committed to the public repository as commit `0d3a057` at 16:49:32 UTC on 9 October 2026, after the W0 results of the registered analysis were known and before the study extracted the July–September logs. It fixed rule B with the margin $\delta = 0.01$, the half-width of the W0 interval of the F2 contrast in the registered analysis (0.0040), rounded up. The extraction manifest records no use of the testing override (Appendix 8.1). The July–September logs held 6,805 GMX V2 closes by contract accounts, all of which decoded. The revised scores were evaluated afterwards, with 400 paired resamples and seed 42, and rule B and R1–R3 are applied to them as a re-analysis; the registered analysis reached the same verdicts (Appendix 8.6).

The spender cohort is that of Section 4.6.2. Seventy GMX V2 contract traders in the freeze-date graph closed at least three positions in the label window; 9 of them had at least one liquidation, and 1 had every close liquidated. Table 4.15 gives every score of rule B on its three labels, and Table 4.16 the contrasts with their verdicts.

The rule is not met. F1 holds: C-PR ranks the spenders that later gained new approval pairs at $\tau = 0.250$ ($[0.240, 0.260]$) against 0.141 ($[0.129, 0.154]$) for its transfer layer walked alone, a difference of $+0.110$ (interval $[0.098, 0.122]$), so the W0 gain on authorization recurs. The in-approve degree at the freeze ranks this label at 0.268 ($[0.257, 0.279]$), above C-PR's point estimate, where in W0 the two point estimates were level; no paired contrast of C-PR with the degree was computed in either window, so F1 shows a gain over the transfer walk, part of which may come from the spenders it scores zero, not a gain over the approval count. F2 fails. On new transfer senders C-PR reaches 0.263 ($[0.254, 0.273]$) against 0.291 ($[0.282, 0.301]$); the difference is -0.028 ($[-0.033, -0.023]$), and the whole interval lies below the margin of -0.01. F3 fails narrowly. On the liquidation-free close rate of the traders C-PR and the transfer walk both reach 0.350 ($[0.236, 0.461]$), and the difference is $+0.000$ ($[-0.014, 0.010]$), whose lower end lies below -0.01. The rule needed all three hypotheses, so, as the registration said for this case, the thesis makes no claim that C-PR

Table 4.15: Window W1: Kendall τ between scores frozen at 30 June 2026 and the three labels of rule B, counted from 1 July 2026 to 30 September 2026. The spenders are the $n = 18{,}152$ contract spenders holding a positive latest allowance at the freeze; 1,616 of them gained a new approval pair and 1,533 a new sender. The traders are the $n = 70$ GMX V2 contract accounts in the freeze-date graph with at least three closes in the label window; 9 of them had at least one liquidation and 1 had only liquidations. C-PR ($\lambda = 0$) is the comparator of rule B and AWP that of its first sensitivity analysis (Table 4.16); EndorseRank is in Table 4.13. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Spenders ($n = 18{,}152$)		Traders ($n = 70$)
Score at 30 June 2026	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.268 [0.257, 0.279]	0.205 [0.193, 0.215]	0.394 [0.261, 0.524]
Transfer in-degree at the freeze (baseline)	0.131 [0.119, 0.143]	0.308 [0.299, 0.319]	0.310 [0.185, 0.416]
AWP	0.169 [0.157, 0.183]	0.330 [0.321, 0.340]	0.329 [0.215, 0.439]
C-PR ($\lambda = 1$)	0.228 [0.217, 0.238]	0.188 [0.178, 0.198]	0.301 [0.177, 0.425]
C-PR ($\lambda = 0$)	0.141 [0.129, 0.154]	0.291 [0.282, 0.301]	0.350 [0.236, 0.461]
C-PR ($\lambda = 0.25$)	0.251 [0.241, 0.260]	0.270 [0.260, 0.279]	0.349 [0.234, 0.459]
C-PR ($\lambda = 0.5$)	0.250 [0.240, 0.260]	0.263 [0.254, 0.273]	0.350 [0.236, 0.461]
C-PR ($\lambda = 0.75$)	0.249 [0.238, 0.259]	0.256 [0.246, 0.266]	0.347 [0.231, 0.462]
S-PR	0.155 [0.142, 0.168]	0.291 [0.281, 0.301]	0.374 [0.255, 0.498]

improves on a walk over the transfers alone.

The registration warned that F3 could pass with both coefficients near zero. Neither is near zero here: every score of Table 4.15 ranks this label with an interval above zero, from 0.301 ($[0.177, 0.425]$) for C-PR ($\lambda = 1$) to 0.394 ($[0.261, 0.524]$) for the in-approve degree at the freeze, the highest point estimate; of the walks only EndorseRank was contrasted with it (Section 4.8).

Of the results reported beside the decision, F4 fails: on the liquidation-free close rate C-PR is not better than the transfer walk. F5 fails, since C-PR and the allowance walk cannot be told apart on that label ($+0.049$, $[-0.055, 0.168]$). F6, the primary part of rule A, fails again: C-PR lies above the allowance walk on new approval pairs ($+0.022$, $[0.015, 0.029]$), which meets the no-worse condition, but below the transfer walk on new senders. As in W0, C-PR's coefficients are flat across λ on new approval pairs and fall on new senders as λ rises. S-PR ranks new senders at 0.291 ($[0.281, 0.301]$), level with the transfer walk, and new approval pairs at 0.155 ($[0.142, 0.168]$), below the allowance walk (0.228, $[0.217, 0.238]$); no paired contrast of S-PR was registered for W1.

Table 4.16: Window W1: the paired contrasts F1–F6 of rule B with their verdicts, and the two sensitivity analyses (rule B and its sensitivity analyses were registered for the raw-unit analysis; here they are applied to the revised scores). $\Delta\tau = \tau(a) - \tau(b)$, with C-PR ($\lambda = 0.5$) as a. Superiority holds when the interval lies above zero; non-inferiority holds when its lower end lies above $-\delta = -0.01$; F6a, the first part of rule A, holds when the interval contains zero or lies above it. F1, F2, F6a and F6b are on the spenders ($n = 18{,}152$ contracts), F3–F5 on the traders ($n = 70$), as in Table 4.15. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Contrast	Label	Test	$\tau(a)$	$\tau(b)$	$\Delta\tau$ [95% CI]	Verdict
Rule B: C-PR ($\lambda = 0.5$) against its transfer layer walked alone, C-PR ($\lambda = 0$)							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.250	0.141	+0.110 [0.098, 0.122]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.263	0.291	-0.028 [-0.033, -0.023]	fails
F3	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	non-inferiority, -0.01	0.350	0.350	+0.000 [-0.014, 0.010]	fails
Decision: F1, F2 and F3 must all hold							not met
Reported, not part of the decision							
F4	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	superiority	0.350	0.350	+0.000 [-0.014, 0.010]	fails
F5	C-PR vs. C-PR ($\lambda = 1$)	liquidation-free close rate	superiority	0.350	0.301	+0.049 [-0.055, 0.168]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.250	0.228	+0.022 [0.015, 0.029]	holds
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.263	0.291	-0.028 [-0.033, -0.023]	fails
Sensitivity analysis 1: AWP as the comparator							
F1	C-PR vs. AWP	new approval pairs	superiority	0.250	0.169	+0.081 [0.069, 0.095]	holds
F2	C-PR vs. AWP	new transfer senders	non-inferiority, -0.01	0.263	0.330	-0.067 [-0.072, -0.061]	fails
F3	C-PR vs. AWP	liquidation-free close rate	non-inferiority, -0.01	0.350	0.329	+0.021 [-0.019, 0.060]	fails
F4	C-PR vs. AWP	liquidation-free close rate	superiority	0.350	0.329	+0.021 [-0.019, 0.060]	fails
Sensitivity analysis 2: contracts outside a graph kept as isolated nodes							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.250	0.141	+0.109 [0.098, 0.121]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.263	0.291	-0.028 [-0.033, -0.023]	fails
F3	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	non-inferiority, -0.01	0.350	0.350	+0.000 [-0.014, 0.010]	fails
F4	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	superiority	0.350	0.350	+0.000 [-0.014, 0.010]	fails
F5	C-PR vs. C-PR ($\lambda = 1$)	liquidation-free close rate	superiority	0.350	0.322	+0.028 [-0.047, 0.094]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.250	0.228	+0.022 [0.015, 0.029]	holds
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.263	0.291	-0.028 [-0.033, -0.023]	fails

Neither registered sensitivity analysis changes the verdict (Table 4.16). With AWP as the comparator, F1 still holds ($+0.081$, $[0.069, 0.095]$), a lead that comes from the spenders AWP scores zero (Section 4.7.4), and F2 fails clearly: AWP ranks new senders at 0.330 ($[0.321, 0.340]$), and C-PR's difference to it is -0.067 ($[-0.072, -0.061]$), an interval that lies entirely below the margin. F3 and F4 fail ($+0.021$, $[-0.019, 0.060]$). On 70 traders this interval is 0.079 wide, so F3 could have held only with a lead of about 0.03; its failure leaves a loss of 0.01 possible and does not show that C-PR ranks the label worse. With the contracts outside a graph kept as isolated nodes, every verdict stays as it was. The point estimates change by at most 0.001, except on F5: the rule lifts the traders outside the allowance graph from zero to the score of a node without in-edges, the allowance walk's coefficient on the liquidation-free close rate rises from 0.301 to 0.322, and C-PR's lead over it shrinks to $+0.028$ ($[-0.047, 0.094]$).

Table 4.17 gives the W1 trader labels that do not enter the decision, with the 2,000 resamples of Section 4.8. On the two count labels the walks that use transfer edges again lead, while the transfer in-degree, near zero in W0, now ranks both with intervals above zero (0.185, $[0.014, 0.341]$; 0.168, $[0.028, 0.291]$), and the intervals of C-PR at $\lambda = 0.5$ and 0.75 on profitable closes contain zero. AWP has the highest point estimates on profitable closes (0.288, $[0.125, 0.442]$) and realized gain (0.346, $[0.189, 0.490]$); its lead over C-PR ($\lambda = 0$) is resolved on the first (C-PR ($\lambda = 0$) minus AWP -0.088, $[-0.148, -0.030]$) but not on the second (-0.008, $[-0.072, 0.061]$). The allowance-side scores stay near zero on both. On the no-liquidation flag every score is positive, the in-approve degree highest (0.407, $[0.268, 0.541]$). On both shares the intervals of the in-approve degree, C-PR ($\lambda = 1$), EndorseRank and its variant with activity restarts lie below zero, for the in-approve degree at -0.294 ($[-0.468, -0.106]$) on the non-loss share, while AWP and C-PR ($\lambda = 0$) are positive with intervals that contain zero.

4.7.4 C-PR beside every other score

Rules A and B leave open how C-PR stands against the other scores of the study. Table 4.18 sets it against each of them and against the two raw degrees, on the spender labels of both windows

Table 4.17: Window W1, traders: Kendall τ between scores frozen at 30 June 2026 and the other GMX V2 labels counted from 1 July 2026 to 30 September 2026, on the $n = 70$ contract accounts with at least three closes in that window. No liquidation is 1 when none of the account's closes was a liquidation. Profitable closes and realized gain grow with the number of closes; the two shares do not. These labels are reported and do not enter rule B. The intervals use the resamples of Table 4.19. CI = 95% percentile bootstrap over contracts (2,000 paired resamples, seed 42).

Score at 30 June 2026	No liquidation	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at the freeze (baseline)	0.407 [0.268, 0.541]	0.042 [-0.158, 0.243]	0.033 [-0.164, 0.230]	-0.221 [-0.406, -0.034]	-0.294 [-0.468, -0.106]
Transfer in-degree at the freeze (baseline)	0.329 [0.189, 0.451]	0.185 [0.014, 0.341]	0.168 [0.028, 0.291]	-0.025 [-0.194, 0.131]	-0.042 [-0.203, 0.114]
EndorseRank	0.303 [0.172, 0.422]	0.073 [-0.115, 0.260]	0.000 [-0.198, 0.200]	-0.180 [-0.340, -0.012]	-0.264 [-0.408, -0.106]
EndorseRank, restarts by approving activity	0.294 [0.158, 0.417]	0.057 [-0.123, 0.239]	-0.004 [-0.179, 0.181]	-0.199 [-0.346, -0.039]	-0.269 [-0.412, -0.118]
AWP	0.333 [0.207, 0.447]	0.288 [0.125, 0.442]	0.346 [0.189, 0.490]	0.136 [-0.039, 0.300]	0.082 [-0.077, 0.247]
C-PR ($\lambda = 1$)	0.307 [0.174, 0.428]	0.010 [-0.167, 0.192]	-0.066 [-0.249, 0.121]	-0.208 [-0.370, -0.030]	-0.287 [-0.443, -0.121]
C-PR ($\lambda = 0$)	0.358 [0.228, 0.475]	0.199 [0.043, 0.355]	0.338 [0.178, 0.485]	0.136 [-0.034, 0.290]	0.038 [-0.127, 0.201]
C-PR ($\lambda = 0.25$)	0.357 [0.227, 0.474]	0.155 [0.002, 0.305]	0.284 [0.122, 0.435]	0.090 [-0.075, 0.242]	-0.011 [-0.173, 0.148]
C-PR ($\lambda = 0.5$)	0.358 [0.229, 0.475]	0.114 [-0.044, 0.267]	0.240 [0.075, 0.397]	0.053 [-0.115, 0.212]	-0.053 [-0.217, 0.103]
C-PR ($\lambda = 0.75$)	0.353 [0.226, 0.472]	0.081 [-0.078, 0.244]	0.201 [0.023, 0.368]	0.016 [-0.153, 0.182]	-0.090 [-0.256, 0.072]
S-PR	0.381 [0.248, 0.495]	0.167 [0.018, 0.312]	0.286 [0.122, 0.441]	0.100 [-0.048, 0.244]	-0.011 [-0.164, 0.151]

and on the liquidation-free close rate of the W1 traders, with the resamples of the rules. Only the cells marked in the table belong to a rule; every other contrast was computed after the labels were known and decides nothing.

Table 4.18: Paired contrasts of C-PR ($\lambda = 0.5$) with every other score of this study and with the two raw degrees, on the spender labels of W0 ($n = 17{,}050$ contracts) and W1 ($n = 18{,}152$) and on the liquidation-free close rate of the W1 traders ($n = 70$). Each cell is $\Delta\tau = \tau_{\text{C-PR}} - \tau_b$ over its 95% interval; within a cohort every cell uses the same resamples. Cells marked [A] are the contrasts of rule A, fixed in the analysis plan; [B] marks those named in the registration of W1 (rule B with F5 and F6a) and [S] those of its first sensitivity analysis, with AWP as the comparator. These rules were fixed for the raw-unit analysis and are applied here to the revised scores. Every other cell was computed after the labels were known and belongs to no rule. The lower block repeats the contrasts with AWP and the in-approve degree on the spenders that had received a transfer from another address before the freeze, the spenders AWP can order; that restriction was also chosen after the results were known. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	W0 spenders		W1 spenders		W1 traders
C-PR minus	New approval pairs	New transfer senders	New approval pairs	New transfer senders	Liquidation-free close rate
C-PR ($\lambda = 1$)	+0.029[A] [0.022, 0.036]	+0.074 [0.066, 0.082]	+0.022[B] [0.015, 0.029]	+0.075 [0.066, 0.085]	+0.049[B] [-0.055, 0.168]
C-PR ($\lambda = 0$)	+0.092 [0.080, 0.105]	-0.034[A] [-0.040, -0.029]	+0.110[B] [0.098, 0.122]	-0.028[B] [-0.033, -0.023]	+0.000[B] [-0.014, 0.010]
EndorseRank	+0.021 [0.013, 0.028]	+0.062 [0.054, 0.070]	+0.016 [0.010, 0.024]	+0.069 [0.061, 0.078]	+0.053 [-0.051, 0.171]
EndorseRank (activity restarts)	-0.057 [-0.065, -0.049]	-0.008 [-0.017, 0.001]	-0.044 [-0.051, -0.038]	+0.021 [0.012, 0.030]	+0.062 [-0.039, 0.164]
AWP	+0.056 [0.042, 0.070]	-0.085 [-0.092, -0.078]	+0.081[S] [0.069, 0.095]	-0.067[S] [-0.072, -0.061]	+0.021[S] [-0.019, 0.060]
S-PR	+0.080 [0.069, 0.093]	-0.033 [-0.038, -0.029]	+0.095 [0.085, 0.107]	-0.027 [-0.031, -0.024]	-0.024 [-0.092, 0.034]
In-approve degree at the freeze	+0.002 [-0.007, 0.011]	+0.082 [0.072, 0.091]	-0.017 [-0.024, -0.010]	+0.059 [0.051, 0.066]	-0.044 [-0.130, 0.038]
Transfer in-degree at the freeze	+0.116 [0.104, 0.130]	-0.042 [-0.049, -0.035]	+0.119 [0.108, 0.132]	-0.045 [-0.051, -0.039]	+0.040 [-0.029, 0.124]
Spenders that had received a transfer before the freeze (W0 $n = 14{,}094$, W1 $n = 15{,}059$)					
AWP	-0.037 [-0.045, -0.028]	-0.073 [-0.079, -0.067]	-0.017 [-0.025, -0.009]	-0.056 [-0.061, -0.050]	—
In-approve degree at the freeze	+0.011 [0.001, 0.021]	—	-0.012 [-0.020, -0.004]	—	—

Against EndorseRank, the walk over allowances alone, C-PR ranks both spender labels higher in both windows: new approval pairs by 0.021 ($[0.013, 0.028]$) in W0 and 0.016 ($[0.010, 0.024]$) in W1, and new senders by 0.062 ($[0.054, 0.070]$) and 0.069 ($[0.061, 0.078]$). On the W1 traders it cannot be told apart from EndorseRank ($+0.053$, $[-0.051, 0.171]$). The variant of EndorseRank with activity restarts ranks new approval pairs higher than C-PR in both windows (-0.057, $[-0.065, -0.049]$,

and -0.044, $[-0.051, -0.038]$); on new senders the two are level in W0 (-0.008, $[-0.017, 0.001]$) and C-PR leads in W1 ($+0.021$, $[0.012, 0.030]$). Within the window C-PR leads the variant on all three families (0.365, 0.477 and 0.142 against 0.164, 0.426 and 0.049; Tables 4.3 and 4.8).

Against AWP, the walk over transfers alone, C-PR ranks new approval pairs higher ($+0.056$, $[0.042, 0.070]$, in W0 and $+0.081$, $[0.069, 0.095]$, in W1) and new senders lower (-0.085, $[-0.092, -0.078]$, and -0.067, $[-0.072, -0.061]$). Its lead on approvals comes from the spenders that AWP cannot order. AWP scores zero every spender without a transfer edge up to the freeze, 2,891 of them in W0, and those spenders gain new approval pairs slightly more often than the rest (Section 4.6). Among the spenders that had received a transfer from another address before the freeze, AWP ranks both labels above C-PR in both windows, new approval pairs by 0.037 ($[0.028, 0.045]$) in W0 and 0.017 ($[0.009, 0.025]$) in W1 and new senders by 0.073 ($[0.067, 0.079]$) and 0.056 ($[0.050, 0.061]$) (lower block of Table 4.18; this restriction, too, was chosen after the results were known). Out of window, what C-PR adds to a walk over transfers is an order among the contracts that this walk leaves tied, at the price of a worse order among those it can rank.

Against the raw degrees, C-PR does not rank either label above the count of the same edge type. On new approval pairs it is level with the in-approve degree in W0 ($+0.002$, $[-0.007, 0.011]$) and below it in W1 (-0.017, $[-0.024, -0.010]$); on new senders it is below the transfer in-degree in both windows (-0.042, $[-0.049, -0.035]$, and -0.045, $[-0.051, -0.039]$). Against its ablation S-PR it ranks new approval pairs higher ($+0.080$, $[0.069, 0.093]$, and $+0.095$, $[0.085, 0.107]$) and new senders lower (-0.033, $[-0.038, -0.029]$, and -0.027, $[-0.031, -0.024]$). On the liquidation-free close rate of the 70 traders no contrast of C-PR excludes zero.

Taken together, C-PR ranks every spender label above EndorseRank in both windows, and it ranks later approvals above AWP only through the contracts that AWP leaves tied. It stays below EndorseRank with activity restarts on later approvals, below AWP and the transfer in-degree on later senders, and below the in-approve degree on later approvals in W1. None of these contrasts was fixed in advance.

4.8 Outcomes built from neither edge type

The GMX trader labels are the only outcomes in these data built from neither edge type. The comparison of Section 3.7.3 was registered with rule B, before the study extracted the July–September logs and before it scored the liquidation labels of W0. Here its contrasts, rules and wording are applied to the revised scores as a re-analysis, with W1 as the primary window and W0 as the secondary one. Intervals come from 2,000 paired resamples and are 95% unless marked 97.5%, the Bonferroni interval of the decision contrasts P and Q. Table 4.19 gives the contrasts on the liquidation-free close rate, Table 4.20 the traders that receive approvals, and Table 8.11 (Appendix 8.5.4) all six trader labels.

R1 does not hold. In W1 the in-approve degree at the freeze reaches $\tau = 0.394$ ($[0.265, 0.518]$) on the liquidation-free close rate and the transfer in-degree 0.310 ($[0.184, 0.425]$), so P, their difference, is $+0.084$ (97.5% interval $[-0.026, 0.188]$), an interval that includes zero. Stratified by the number of closes, P is $+0.101$ ($[-0.024, 0.231]$). In W0, P is $+0.182$ (97.5% interval $[0.032, 0.338]$), positive, but R1 also needed both W1 intervals above zero; in the registered wording, there is no count-level lead of allowance over transfer on later liquidation avoidance. R2 does not hold either. Q, EndorseRank with AWP's activity restarts minus AWP (the like-for-like PageRank pair), is -0.041 in W1 and -0.014 in W0, and both 97.5% intervals include zero ($[-0.148, 0.064]$ and $[-0.161, 0.151]$), so the PageRank versions are not resolved. Neither are C-PR's layers (-0.049, $[-0.169, 0.060]$, in W1), EndorseRank against AWP (-0.032, $[-0.133, 0.066]$) or C-PR ($\lambda = 0$) against AWP ($+0.021$, $[-0.016, 0.065]$), in either window. In a contrast the registration did not name, EndorseRank falls below the approval count, by 0.097 ($[-0.174, -0.009]$) in W1 and 0.120 ($[-0.203, -0.033]$) in W0.

The other labels do not enter the decision (Table 8.11). On the no-liquidation flag, P is $+0.078$ ($[-0.012, 0.166]$) in W1 and $+0.197$ ($[0.059, 0.333]$) in W0. On profitable closes and realized gain, which grow with activity, the PageRank pairs favor the transfer side in W1, with intervals below zero for Q (-0.231, $[-0.429, -0.027]$, and -0.350, $[-0.536, -0.157]$) and for EndorseR-

Table 4.19: Allowance-side against transfer-side scores on the liquidation-free close rate of the GMX V2 contract traders with at least three closes in the label window, a label built from neither edge type ($n = 70$ traders in W1, $n = 88$ in W0). The pairs, the decision contrasts P and Q with their Bonferroni 97.5% intervals and the stratified rows were registered before the study extracted the July–September logs, for the raw-unit analysis, and are applied here to the revised scores; W1 is the primary window and W0 the secondary one. Paired contract bootstrap with 2,000 resamples (seed 42). The stratified rows combine Kendall's τ_b within strata of the number of label-window closes (3–4, 5–9, 10–24, 25 or more), weighted by the number of contract pairs. Registered rules applied to the revised scores: R1 (count-level lead) not met; R2 (PageRank-level lead) not met; R3 (specificity) applies.

Contrast (a minus b)	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI	97.5% CI
Window W1, July–September 2026 ($n = 70$ traders)					
In-approve degree minus transfer in-degree	0.394	0.310	0.084	[-0.013, 0.176]	[-0.026, 0.188]
EndorseRank (activity restarts) minus AWP	0.288	0.329	-0.041	[-0.135, 0.050]	[-0.148, 0.064]
C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$)	0.301	0.350	-0.049	[-0.169, 0.060]	–
EndorseRank minus AWP	0.297	0.329	-0.032	[-0.133, 0.066]	–
C-PR ($\lambda = 0$) minus AWP	0.350	0.329	0.021	[-0.016, 0.065]	–
EndorseRank minus in-approve degree	0.297	0.394	-0.097	[-0.174, -0.009]	–
In-approve degree minus transfer in-degree, stratified by closes	0.381	0.280	0.101	[-0.024, 0.231]	–
EndorseRank (activity restarts) minus AWP, stratified by closes	0.273	0.301	-0.028	[-0.207, 0.098]	–
Window W0, April–June 2026 ($n = 88$ traders)					
In-approve degree minus transfer in-degree	0.344	0.161	0.182	[0.052, 0.314]	[0.032, 0.338]
EndorseRank (activity restarts) minus AWP	0.227	0.240	-0.014	[-0.147, 0.130]	[-0.161, 0.151]
C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$)	0.225	0.265	-0.041	[-0.187, 0.105]	–
EndorseRank minus AWP	0.224	0.240	-0.017	[-0.156, 0.128]	–
C-PR ($\lambda = 0$) minus AWP	0.265	0.240	0.025	[-0.022, 0.077]	–
EndorseRank minus in-approve degree	0.224	0.344	-0.120	[-0.203, -0.033]	–
In-approve degree minus transfer in-degree, stratified by closes	0.338	0.209	0.129	[0.020, 0.293]	–
EndorseRank (activity restarts) minus AWP, stratified by closes	0.215	0.252	-0.037	[-0.173, 0.125]	–

ank against AWP (-0.214, $[-0.423, -0.012]$, and -0.346, $[-0.547, -0.144]$); P points the same way without being resolved. R3 applies. In W1 the in-approve degree ranks below the transfer in-degree on the profitable share (-0.196, $[-0.364, -0.046]$) and the non-loss share (-0.252, $[-0.391, -0.120]$), so, in the registered wording, the signal concerns forced-liquidation avoidance, not trading success. Part of the non-loss gap is built in: the pipeline counts a liquidated close as a close that was not a loss, so a trader with liquidations gains non-loss share, and R3 therefore rests mainly on the profitable share, which counts a liquidated close as not profitable.

Table 4.20: GMX V2 contract traders with a positive in-approve degree at the freeze (recipients) against all other labelled traders, in both windows ($n = 70$ traders in W1, $n = 88$ in W0). The comparison of recipients with the other traders and the influence check were registered with the contrasts of Table 4.19 and are applied here to the revised scores. The highest in-approve degree among the traders is 1 in W1 and 3 in W0, so the ten recipients left out by the influence check are taken from tied values in node order. The influence check is a paired contract bootstrap on the remaining traders (2,000 resamples, seed 42).

	Recipients	Other traders
Window W1, July–September 2026 ($n = 70$ traders)		
Traders	37	33
Share with no liquidation	1.000	0.727
Mean liquidation-free close rate	1.000	0.878
Mean profitable share	0.325	0.479
Mean non-loss share	0.433	0.644
Median closes in the label window	24.0	11.0
P without the ten recipients of highest in-approve degree: $\Delta\tau = 0.073$ [-0.039, 0.186]		
Window W0, April–June 2026 ($n = 88$ traders)		
Traders	40	48
Share with no liquidation	1.000	0.750
Mean liquidation-free close rate	1.000	0.919
Mean profitable share	0.362	0.474
Mean non-loss share	0.402	0.607
Median closes in the label window	31.0	10.0
P without the ten recipients of highest in-approve degree: $\Delta\tau = 0.188$ [0.046, 0.331]		

The count-level comparison rests on few traders. In W1 the highest in-approve degree among the traders is 1, so P in effect compares the 37 recipients with the 33 other traders; in W0 the maximum is 3, and 40 of the 88 traders are recipients. Every recipient avoided liquidation in both windows

(37 of 37 and 40 of 40), against 72.7% of the other traders in W1 and 75.0% in W0. Recipients also closed more positions (median 24 against 11 in W1, 31 against 10 in W0) and had lower profitable and non-loss shares (mean non-loss share 0.433 against 0.644 in W1), the pattern behind R3. Without the 10 recipients of highest in-approve degree, P is $+0.073$ ($[-0.039, 0.186]$) in W1 and $+0.188$ ($[0.046, 0.331]$) in W0.

4.9 Claims (C1–C6)

The chapter's results support the claims below, cross-referenced to the hypothesis and research question columns of Table 1.2. C1–C5 are the primary claims; C2 and C3 rest on contrasts computed after the labels were known, and C6 on the comparison on outcomes built from neither edge type. The verdicts of rules A and B in C1 and of R1–R3 in C6 apply the registered rules to the revised scores, as a re-analysis; the registered analysis reached the same verdicts (Appendix 8.6).

C1. Under its rules, C-PR, a fixed-λ mixture of two dollar-weighted layers with uniform restarts, does not beat its two layers walked alone. Rule A fails: C-PR is no worse than the allowance walk on new approval pairs ($+0.029$, $[0.022, 0.036]$) but worse than the transfer walk on new senders (-0.034, $[-0.040, -0.029]$), and its same-window part fails as well (Tables 4.10 and 4.14). Rule B is not met (Table 4.16): the W0 gain over the transfer walk on new approval pairs, found after the labels were known, recurs in W1 (F1, $+0.110$, $[0.098, 0.122]$), but C-PR falls below the transfer walk on new senders by more than the margin of 0.01 (F2, -0.028, $[-0.033, -0.023]$), and F3 fails narrowly ($+0.000$, $[-0.014, 0.010]$). Against AWP, in the first sensitivity analysis, F2 fails clearly (-0.067, $[-0.072, -0.061]$). S-PR is level with the transfer walk on new senders and below the allowance walk on new approval pairs, so allowance edges help mainly when they carry the walk; Section 4.7.1 reports where the hybrids stand within the window.

C2. Beside the rules, C-PR ranks both spender labels above EndorseRank in both windows (new approval pairs $+0.021$, $[0.013, 0.028]$, and $+0.016$, $[0.010, 0.024]$; new senders $+0.062$,

$[0.054, 0.070]$, and $+0.069$, $[0.061, 0.078]$) and leads it within the window on the transfer and Sybil-stability families. It ranks new approval pairs above AWP ($+0.056$, $[0.042, 0.070]$; $+0.081$, $[0.069, 0.095]$) only through the spenders that AWP scores zero: among the spenders that had received a transfer, AWP ranks both labels higher. C-PR ranks new senders below AWP and below the transfer in-degree in both windows, new approval pairs below EndorseRank with activity restarts in both windows (-0.057, $[-0.065, -0.049]$; -0.044, $[-0.051, -0.038]$) and below the in-approve degree in W1 (-0.017, $[-0.024, -0.010]$), and none of its contrasts on the W1 traders excludes zero (Table 4.18). None of these contrasts was fixed in advance.

C3. Out of window the two single-layer walks differ against the raw degrees they smooth. AWP ranks later new transfer senders above the transfer in-degree in both windows (W0 $+0.043$, $[0.038, 0.048]$; W1 $+0.022$, $[0.017, 0.027]$), while EndorseRank ranks later new approval pairs below the in-approve degree in both (W0 -0.019, $[-0.026, -0.011]$; W1 -0.034, $[-0.038, -0.029]$) (Tables 4.10 and 4.13); in W0 neither comparison is a matter of coverage (Table 4.11), and no such restriction was computed for W1. With activity restarts EndorseRank also beats its count (Section 4.5.4), so the restart rule, not the edge type, appears to decide which walk does. On the profit labels of the contract traders of W0 the paired contrasts separate the two sides only on realized gain, where the transfer layer walked alone ranks above the allowance layer (C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$) -0.248, $[-0.433, -0.067]$), not on profitable closes (EndorseRank minus AWP -0.005, $[-0.163, 0.155]$), and no score ranks the share of closes that were profitable (Tables 4.12 and 8.11). These comparisons were computed after the labels were known.

C4. In the same window each single-layer walk agrees most with the proxies of the edge type it walks on, EndorseRank with the allowance family (0.585, $[0.578, 0.592]$) and AWP with the transfer family (0.572, $[0.564, 0.580]$), and the mixtures of C-PR lie between their two layers (Table 4.3). These tests measure internal consistency, and part of the agreement is built in; external validity is outside their scope. The two walks order the cohort differently: their

agreement, $\tau_b = 0.174$ ($[0.163, 0.185]$), has an interval that excludes zero and one; 96.7% of the contract pairs are ordered by both scores, and 41.2% of those are discordant. Neither result comes from ties or from the rule for contracts outside a graph (Table 4.4, Figure 4.3).

C5. On identical contract sets, the cost of each score's solver call grows with the size of its graph, since each iteration costs time in $|E|$ and $|V|$, and with the iterations its walk needs. The allowance graph has 44.8% of the transfer graph's edges, and AWP takes 2.17 times as long as EndorseRank at an edge ratio of 2.23 (Table 4.1). On the subsamples the time ratio follows the edge ratio only from 10,000 contracts on, and the stages do not test edge counts beyond those of the cohort graph (Table 4.2). The low cost is no advantage of the algorithm: C-PR and S-PR walk the transfer layer and take longer than AWP, C-PR 1.62 times as long.

C6. On later liquidation avoidance of the GMX V2 contract traders, the one outcome built from neither edge type that bears on counterparty risk, the approval count at the freeze does not lead the transfer count in the primary window W1 (P $= +0.084$, 97.5% interval $[-0.026, 0.188]$), although P is positive in W0 ($+0.182$, 97.5% interval $[0.032, 0.338]$). The PageRank versions are not resolved (Q $= -0.041$, 97.5% interval $[-0.148, 0.064]$). In W1 the approval count ranks traders lower than the transfer count on the shares of closes that were profitable or not losing, so, in the wording of R3, the signal concerns forced-liquidation avoidance, not trading success; in W0 neither difference is resolved (Table 8.11). The samples are small, 70 and 88 contract traders, and every interval is wide (Section 4.8).

The same-window tables and the rank divergence answer RQ1, the benchmark tables RQ2, the holdouts of the single-layer walks with Table 4.18 RQ3, and Section 4.7 RQ4 and RQ5; C6 adds no research question. The direct comparisons between EndorseRank and AWP are in Appendix 8.5 and Table 4.19, and none of them carries a claim of the thesis. Chapter 5 reads these findings against the research questions and objectives of Chapter 1.